

WTF : A (Causal) relationship btw  $x_i$  and  $y_i$ .

## A Quick Review of OLS

A simple reg model w/ one exp variable ( $x_i$ ).

$$y_i = \beta_0 + \beta_1 x_i + u_i \quad i = \text{obs} \quad \text{assuming } E(u_i | x_i) = 0$$

①

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + u_i \\ \rightarrow \bar{y} &= \beta_0 + \beta_1 \bar{x} + \bar{u} \end{aligned}$$

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\text{Cov}(x_i, y_i)}{\text{Var}(x_i)}$$

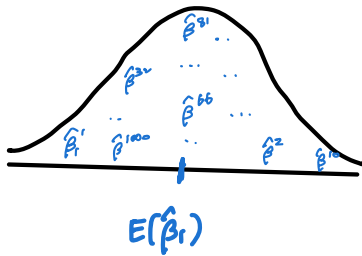
$$y_i - \bar{y} = \beta_1 (x_i - \bar{x}) + (u_i - \bar{u})$$

What does "bias" mean?

From random sampling with a large sample,

if we estimate  $\beta_1$  multiple times, we have  $\hat{\beta}_1^1, \hat{\beta}_1^2, \hat{\beta}_1^3 \dots \hat{\beta}_1^{1000}$  (1,000 times)

If we plot all these values and if it looks like this:



then  $E(\hat{\beta}_1) = \beta_1$ .

On average, estimated  $\hat{\beta}_1$  equals the true parameter  $\beta_1$ .

In this case, we say that  $\hat{\beta}_1$  is unbiased.

To sum up, we say that  $\hat{\beta}_1$  is unbiased if  $E(\hat{\beta}_1) = \beta_1$ .

Is  $\hat{\beta}_1$  from the OLS unbiased?

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \stackrel{①}{=} \frac{\sum (x_i - \bar{x}) [\beta_1 (x_i - \bar{x}) + (u_i - \bar{u})]}{\sum (x_i - \bar{x})^2}$$

$$= \beta_1 + \frac{\sum (x_i - \bar{x})(u_i - \bar{u})}{\sum (x_i - \bar{x})^2}$$

$$E(\hat{\beta}_1) = E \left[ \beta_1 + \frac{\sum (x_i - \bar{x})(u_i - \bar{u})}{\sum (x_i - \bar{x})^2} \right]$$

$$= \beta_1 + E \left[ \frac{\sum (x_i - \bar{x})(u_i - \bar{u})}{\sum (x_i - \bar{x})^2} \right] \quad \text{because } E(u_i | x_i) = 0$$

What if  $E(u_i | x_i) \neq 0$ ?

Then, we have something other than  $\beta_1$ .  
 $\therefore \hat{\beta}_1$  is biased.

## What is Omitted Variable Bias?

Suppose the true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

where  $E[u_i | x_{1i}, x_{2i}] = 0$

If we mistakenly run a regression w/o  $x_2$ ,

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i, \text{ where } u_i = x_{2i} + \varepsilon_i$$

$$\tilde{\beta}_1 = \frac{\sum (x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum (x_{1i} - \bar{x}_1)^2} \stackrel{b)}{=} \frac{\sum (x_{1i} - \bar{x}_1) [\beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + (u_i - \bar{u})]}{\sum (x_{1i} - \bar{x}_1)^2}$$

$$= \beta_1 + \beta_2 \cdot \frac{\sum (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\sum (x_{1i} - \bar{x}_1)^2} + \frac{\sum (x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\sum (x_{1i} - \bar{x}_1)^2}$$

$$E[\tilde{\beta}_1] = E \left[ \beta_1 + \beta_2 \cdot \frac{\sum (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\sum (x_{1i} - \bar{x}_1)^2} + \frac{\sum (x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\sum (x_{1i} - \bar{x}_1)^2} \right]$$

$$= \beta_1 + \beta_2 E \left[ \frac{\sum (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\sum (x_{1i} - \bar{x}_1)^2} \right] + E \left[ \frac{\sum (x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\sum (x_{1i} - \bar{x}_1)^2} \right]$$

$$= \beta_1 + \beta_2 \cdot \frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)}$$

Isn't this familiar?

$$= \beta_1 + \beta_2 \cdot \delta_1$$

Direct rel  
btw  $y$  &  $x_1$

Direct rel  
tw  $x_1$  &  $x_2$

Direct rel  
btw  $y$  &  $x_2$

$$= \beta_1 + \underbrace{\beta_2 \delta_1}_{\text{Bias}}$$

b)

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$$

$$\bar{y} = \beta_0 + \beta_1 \bar{x}_1 + \beta_2 \bar{x}_2 + \bar{u}$$

$$y_i - \bar{y} = \beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + (u_i - \bar{u})$$

• regress  $y$  on  $x$

$$y = \alpha_0 + \alpha_1 x + e$$

$$\alpha_1 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

• what we have :

$$\frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)}$$

⇒ It's the coeff on  $x_1$  from regressing  $x_2$  on  $x_1$

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

To sum up,

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \cdot \delta_1$$

• Definition of Unbiasedness

if  $\hat{\beta}_1$  is unbiased,  $E(\hat{\beta}_1) = \beta_1$ .

$$E(\tilde{\beta}) = \begin{cases} \beta_1 & \text{if } \beta_2 \delta_1 = 0 \text{ ("unbiased")} \Leftrightarrow \beta_2 = 0 \text{ OR } \delta_1 = 0 \\ \beta_1 + \beta_2 \delta_1 & \text{if } \beta_2 \delta_1 \neq 0 \text{ ("biased")} \Leftrightarrow \beta_2 \neq 0 \text{ AND } \delta_1 \neq 0 \end{cases}$$

( $\therefore$ ) we call this term as "BIAS"  
because it biases our estimation.

What are  $\beta_2$  and  $\delta_1$ , again?

$\beta_2$ : Direct rel btw  $y \in \mathcal{X}_2$  from  $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$

$\delta_1$ : Direct rel btw  $x_1 \in \mathcal{X}_2$  from  $x_{2i} = \delta_0 + \delta_1 x_{1i} + v_i$

Because this bias is caused by omitting  $x_2$ , which is related to BOTH  $y$  and  $x_1$ , we call this as Omitted Variable Bias (OVB).

## Main Takeaway

if we run a regression omitting a variable that is BOTH correlated with  $y$  and  $x_1$ , our estimate has an OVB problem.

(We cannot say that  $x$  and  $y$  have a causal relationship.)

the variable of interest

## (Example) LAB4. Question 7

True model :  $\ln E = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + \beta_3 \text{friend4} + \varepsilon$  (7)

What we estimated :  $\ln E = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + u$  (6)  
(in Q6)

$$E(\tilde{\beta}_1) = \beta_1 + \underbrace{\hat{\beta}_3}_{0.05} \cdot \underbrace{\hat{\delta}_1}_{0.082}$$

Bias > 0 "upward bias"

the TRUE parameter  $\beta_1$  from the true model (7)

<Auxiliary regression>

$$\begin{aligned} \text{friend4} &= \delta_0 + \delta_1 \text{indegree} + v \\ &= 4.354 + 0.082 \text{indegree} \end{aligned}$$

What we estimated from Equation (6)

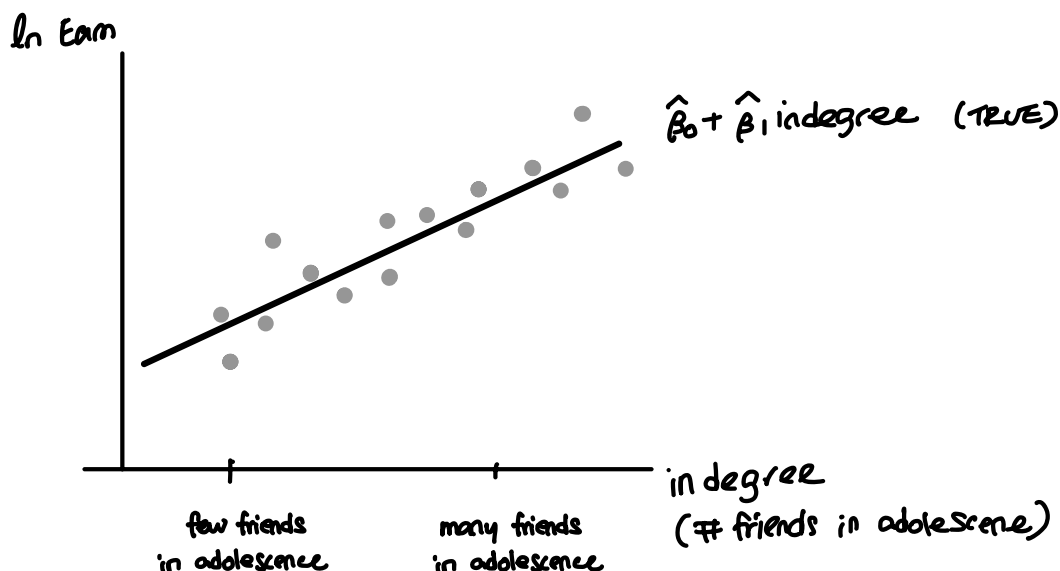
⇒ What can you say about  $\tilde{\beta}_1$  ?

The estimated  $\tilde{\beta}_1$  is larger than the true parameter  $\beta_1$ .

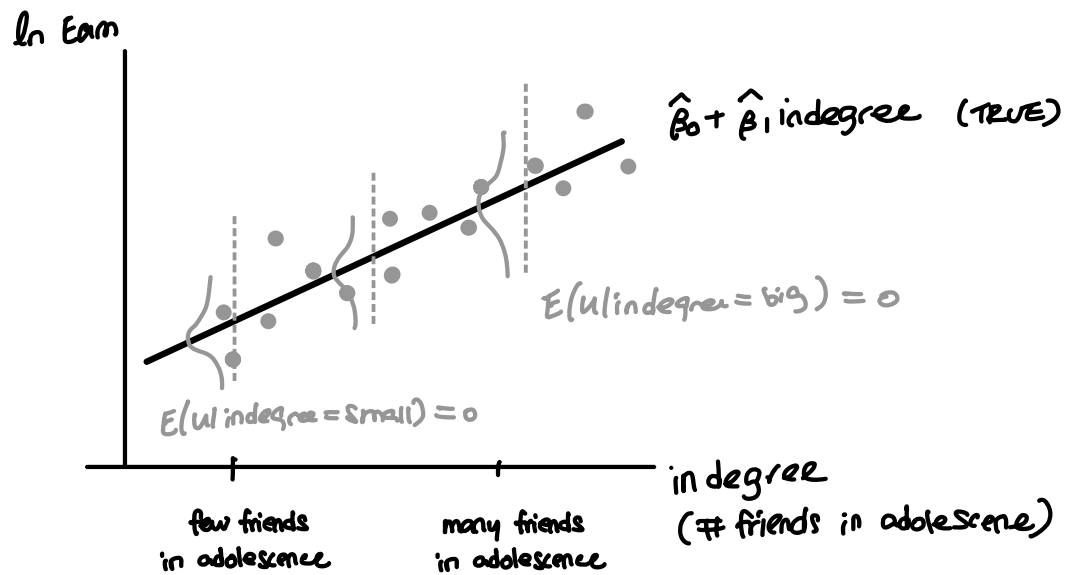
(i.e.) The true parameter is smaller than what we estimated in (6).

We also say that " $\tilde{\beta}_1$  is biased away from zero."

Suppose this is from the true model



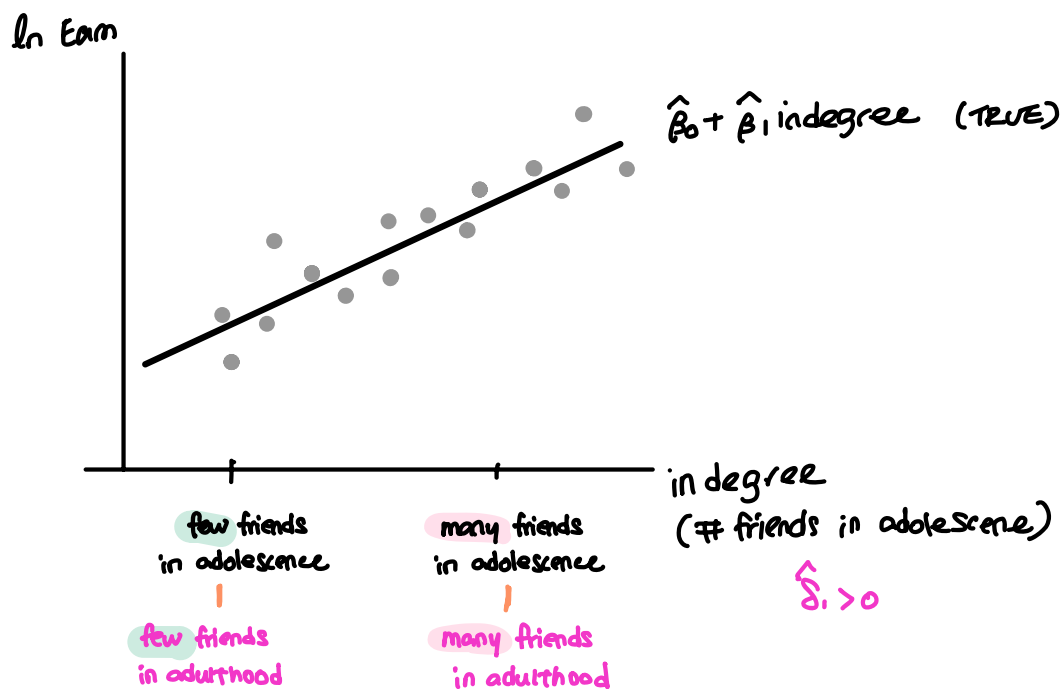
Suppose this is from the true model



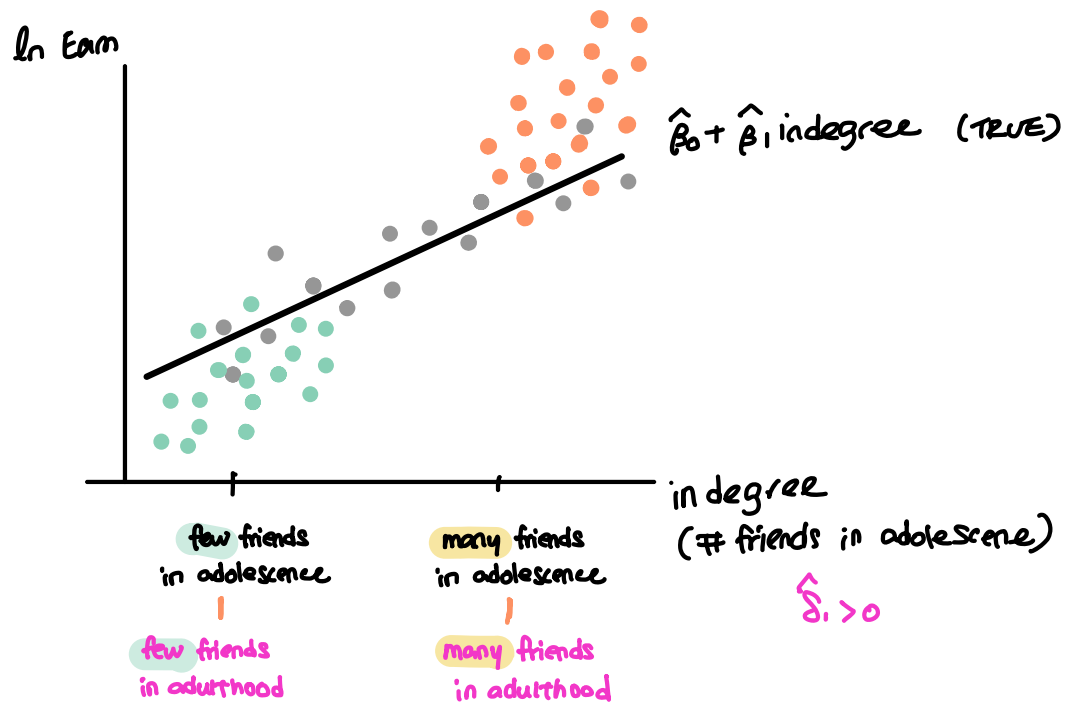
$$u = \text{friend}_4 + \varepsilon$$

However, if we omit 'Friend<sub>4</sub>' in the model, it is now part of the error term.

$\hat{\beta}_1 > 0$  : "A positive rel btw # friends in adolescence & # friends in adulthood."

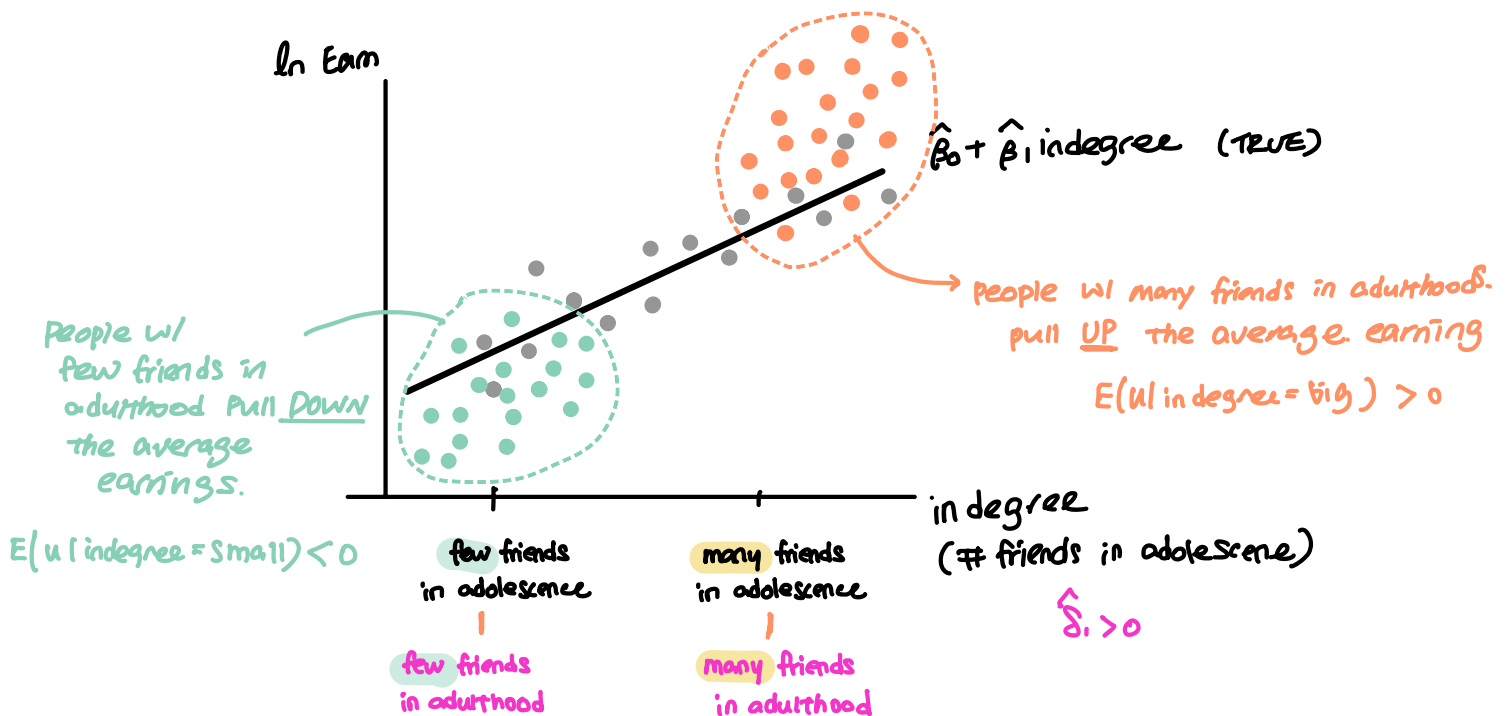


$\hat{\beta}_2 > 0$ : "A positive relationship btw Earnings & # friends in adulthood."

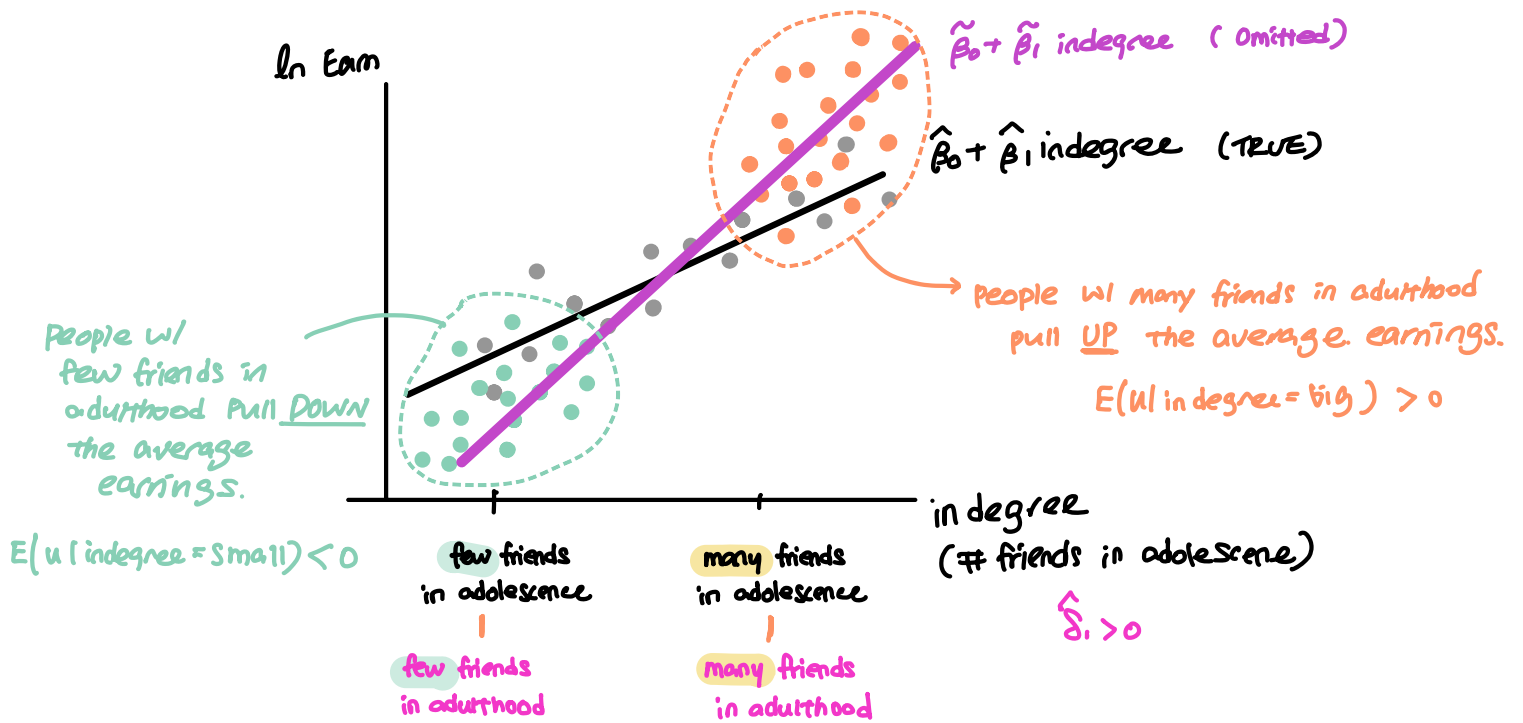


If we omit Friend\_4 in the reg model, the error term  $u = \text{Friend}_4 + \varepsilon$ .

Because Friend\_4 is correlated with  $\ln \text{Earn}$  AND  $\text{in degree}$ ,  $E[u | \text{in-degree}] \neq \phi$ .



Therefore ...



True model :  $\ln \text{Earn} = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + \beta_3 \text{friend}_4 + \varepsilon$

. reg ln\_earn in\_degree female friend\_4

| Source   | SS         | df    | MS         | Number of obs | = | 2,601  |
|----------|------------|-------|------------|---------------|---|--------|
| Model    | 134.501255 | 3     | 44.8337517 | F(3, 2597)    | = | 43.03  |
| Residual | 2705.73506 | 2,597 | 1.04186949 | Prob > F      | = | 0.0000 |
|          |            |       |            | R-squared     | = | 0.0474 |
|          |            |       |            | Adj R-squared | = | 0.0463 |
| Total    | 2840.23631 | 2,600 | 1.09239858 | Root MSE      | = | 1.0207 |

| ln_earn                   | Coefficient | Std. err. | t      | P> t  | [95% conf. interval] |           |
|---------------------------|-------------|-----------|--------|-------|----------------------|-----------|
| $\hat{\beta}_1$ in_degree | .0200568    | .005484   | 3.66   | 0.000 | .0093035             | .0308102  |
| female                    | -.3045655   | .0404073  | -7.54  | 0.000 | -.3837993            | -.2253316 |
| $\hat{\beta}_3$ friend_4  | .0503541    | .0073086  | 6.89   | 0.000 | .0360229             | .0646853  |
| _cons                     | 10.00184    | .0513622  | 194.73 | 0.000 | 9.901127             | 10.10256  |

. reg friend\_4 in\_degree : Friend 4 =  $\delta_0 + \delta_1 \text{in degree} + v$  (Auxiliary regression)

| Source   | SS         | df    | MS         | Number of obs | = | 2,893  |
|----------|------------|-------|------------|---------------|---|--------|
| Model    | 268.635594 | 1     | 268.635594 | F(1, 2891)    | = | 35.14  |
| Residual | 22103.3174 | 2,891 | 7.64556119 | Prob > F      | = | 0.0000 |
|          |            |       |            | R-squared     | = | 0.0120 |
|          |            |       |            | Adj R-squared | = | 0.0117 |
| Total    | 22371.953  | 2,892 | 7.7358067  | Root MSE      | = | 2.7651 |

| friend_4                   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |          |
|----------------------------|-------------|-----------|-------|-------|----------------------|----------|
| $\hat{\delta}_1$ in_degree | .0824532    | .0139101  | 5.93  | 0.000 | .0551785             | .1097279 |
| _cons                      | 4.353976    | .0828171  | 52.57 | 0.000 | 4.191589             | 4.516362 |

**Problem 1.** Use the regression results below to answer the questions. *lprice* log(price, \$1,000), *lsqrtft* log(size of house in square feet), *llotsize* log(size of lot in square feet), *colonial* colonial style (=1), *bdrms* number of bedrooms

```
. reg lprice lsqrtft llotsize colonial bdrms
```

| Source   | SS                | df        | MS                | Number of obs | = | 88            |
|----------|-------------------|-----------|-------------------|---------------|---|---------------|
| Model    | <b>5.20397919</b> | <b>4</b>  | <b>1.3009948</b>  | F(4, 83)      | = | <b>38.38</b>  |
| Residual | <b>2.81362433</b> | <b>83</b> | <b>.033899088</b> | Prob > F      | = | <b>0.0000</b> |
|          |                   |           |                   | R-squared     | = | <b>0.6491</b> |
|          |                   |           |                   | Adj R-squared | = | <b>0.6322</b> |
| Total    | <b>8.01760352</b> | <b>87</b> | <b>.092156362</b> | Root MSE      | = | <b>.18412</b> |

| <i>lprice</i>   | Coefficient      | Std. err.       | t            | P> t         | [95% conf. interval] |                  |
|-----------------|------------------|-----------------|--------------|--------------|----------------------|------------------|
| <i>lsqrtft</i>  | <b>.7071931</b>  | <b>.092802</b>  | <b>7.62</b>  | <b>0.000</b> | <b>.5226138</b>      | <b>.8917725</b>  |
| <i>llotsize</i> | <b>.1678189</b>  | <b>.0381807</b> | <b>4.40</b>  | <b>0.000</b> | <b>.0918791</b>      | <b>.2437587</b>  |
| <i>colonial</i> | <b>.0537962</b>  | <b>.0447732</b> | <b>1.20</b>  | <b>0.233</b> | <b>-.035256</b>      | <b>.1428483</b>  |
| <i>bdrms</i>    | <b>.0268305</b>  | <b>.0287236</b> | <b>0.93</b>  | <b>0.353</b> | <b>-.0302995</b>     | <b>.0839605</b>  |
| _cons           | <b>-1.349589</b> | <b>.651041</b>  | <b>-2.07</b> | <b>0.041</b> | <b>-2.644483</b>     | <b>-.0546947</b> |

(a) Interpret the coefficient on *lsqrtft*

(b) Interpret the coefficient on *colonial*

(c) Interpret the coefficient on *bdrms*

**Problem 2.** Based on the table below, answer the questions.

| Dependent Variable=Labor Force Participation |                    |                    |                    |                    |
|----------------------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                              | (1)                | (2)                | (3)                | (4)                |
|                                              | LPM                | Probit             | LPM                | Probit             |
| HHI                                          | -0.11**<br>(0.006) | -0.38**<br>(0.019) | -0.09**<br>(0.006) | -0.32**<br>(0.021) |
| LF experience                                |                    |                    | -0.01**<br>(0.000) | -0.03**<br>(0.001) |
| nchild<6yrs old                              |                    |                    | -0.14**<br>(0.005) | -0.49**<br>(0.016) |
| nchild 6~18yrs old                           |                    |                    | -0.02**<br>(0.003) | -0.09**<br>(0.010) |
| Husband's earned income<br>(\$10,000)        |                    |                    | -0.01**<br>(0.001) | -0.02**<br>(0.004) |
| Constant                                     | 0.83**<br>(0.004)  | 0.97**<br>(0.016)  | 1.09**<br>(0.008)  | 1.97**<br>(0.039)  |
| Observations                                 | 22,369             | 22,369             | 22,259             | 22,259             |
| R-squared                                    | 0.017              |                    | 0.071              |                    |
| Robust standard errors in parentheses        |                    |                    |                    |                    |
| ** p<0.01, * p<0.05, + p<0.1                 |                    |                    |                    |                    |

- (a) Suppose a married woman is covered by her husband's insurance (HHI), has 10 years of working experience, and 2 children who are younger than 6 years old and her husband's income is \$30,000. In column (3), what is the probability that the woman will participate in the labor force?

- (b) In column (4), what is the probability that the woman will participate in the labor force?

**Problem 3.** Using data from a private school voucher program, also known as a school choice program, we estimated the following regression. *choicelyrs* represents the number of years a student attended a choice school from 1991 to 1994. Since it is a categorical variable that takes the values 0, 1, 2, 3, and 4, we generated binary variables *choicelyrs1*, *choicelyrs2*, *choicelyrs3*, and *choicelyrs4*.

```
. reg mathscore choicelyrs1 choicelyrs2 choicelyrs3 choicelyrs4 black hispanic female
```

| Source   | SS         | df  | MS         | Number of obs | = | 990    |
|----------|------------|-----|------------|---------------|---|--------|
|          |            |     |            | F(7, 982)     | = | 13.52  |
| Model    | 37872.1524 | 7   | 5410.30749 | Prob > F      | = | 0.0000 |
| Residual | 392958.98  | 982 | 400.161894 | R-squared     | = | 0.0879 |
|          |            |     |            | Adj R-squared | = | 0.0814 |
| Total    | 430831.132 | 989 | 435.622985 | Root MSE      | = | 20.004 |

| mathscore   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|-------------|-------------|-----------|-------|-------|----------------------|-----------|
| choicelyrs1 | .5102603    | 1.735531  | 0.29  | 0.769 | -2.895516            | 3.916037  |
| choicelyrs2 | -1.314817   | 2.098575  | -0.63 | 0.531 | -5.433025            | 2.803392  |
| choicelyrs3 | -2.766788   | 2.198078  | -1.26 | 0.208 | -7.080257            | 1.546681  |
| choicelyrs4 | -.617641    | 2.856311  | -0.22 | 0.829 | -6.222817            | 4.987535  |
| black       | -16.19045   | 1.849133  | -8.76 | 0.000 | -19.81916            | -12.56174 |
| hispanic    | -13.37447   | 2.40842   | -5.55 | 0.000 | -18.10072            | -8.648231 |
| female      | 1.370047    | 1.277174  | 1.07  | 0.284 | -1.136258            | 3.876351  |
| _cons       | 57.04585    | 1.660486  | 34.35 | 0.000 | 53.78734             | 60.30436  |

(a) Interpret the coefficient on *choicelyrs1*

(b) How would you rewrite a regression model to examine whether math scores differ between students who have ever attended a choice school and those who have never attended?

**Problem 4.** The following regression results are based on a dataset of working men from 1991, with *abil* representing cognitive ability and *educ* representing years of education.

**. reg educ motheduc fatheduc abil abil2**

| Source   | SS                | df           | MS                | Number of obs | = | 1,230         |
|----------|-------------------|--------------|-------------------|---------------|---|---------------|
| Model    | <b>3027.03706</b> | <b>4</b>     | <b>756.759264</b> | F(4, 1225)    | = | <b>244.91</b> |
| Residual | <b>3785.24262</b> | <b>1,225</b> | <b>3.08999397</b> | Prob > F      | = | <b>0.0000</b> |
|          |                   |              |                   | R-squared     | = | <b>0.4444</b> |
|          |                   |              |                   | Adj R-squared | = | <b>0.4425</b> |
| Total    | <b>6812.27967</b> | <b>1,229</b> | <b>5.54294522</b> | Root MSE      | = | <b>1.7578</b> |

| educ     | Coefficient     | Std. err.       | t            | P> t         | [95% conf. interval] |                 |
|----------|-----------------|-----------------|--------------|--------------|----------------------|-----------------|
| motheduc | <b>.1901261</b> | <b>.0280957</b> | <b>6.77</b>  | <b>0.000</b> | <b>.1350051</b>      | <b>.2452472</b> |
| fatheduc | <b>.1089387</b> | <b>.0196014</b> | <b>5.56</b>  | <b>0.000</b> | <b>.0704827</b>      | <b>.1473946</b> |
| abil     | <b>.4014624</b> | <b>.0302875</b> | <b>13.26</b> | <b>0.000</b> | <b>.3420413</b>      | <b>.4608835</b> |
| abil2    | <b>.050599</b>  | <b>.0083039</b> | <b>6.09</b>  | <b>0.000</b> | <b>.0343076</b>      | <b>.0668905</b> |
| _cons    | <b>8.240226</b> | <b>.2874099</b> | <b>28.67</b> | <b>0.000</b> | <b>7.676356</b>      | <b>8.804097</b> |

- Draw the relationship between cognitive ability (*abil*) and years of education (*educ*).
- Find the turning point.
- Conduct a significance test at the 5% level to evaluate whether the relationship between cognitive ability and years of education is linear.

**Problem 5.** Using the same dataset, we run a different regression to examine the effect of number of siblings and college tuition. *sibs* is the number of siblings and *tuit17* is college tuition at age 17.

```
. gen sib3 = sibs >= 3
```

```
. reg educ c.tuit17##sib3
```

| Source   | SS         | df    | MS         | Number of obs | = | 1,230  |
|----------|------------|-------|------------|---------------|---|--------|
| Model    | 305.529838 | 3     | 101.843279 | F(3, 1226)    | = | 19.19  |
| Residual | 6506.74984 | 1,226 | 5.30730003 | Prob > F      | = | 0.0000 |
|          |            |       |            | R-squared     | = | 0.0448 |
|          |            |       |            | Adj R-squared | = | 0.0425 |
| Total    | 6812.27967 | 1,229 | 5.54294522 | Root MSE      | = | 2.3038 |

| educ          | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|---------------|-------------|-----------|-------|-------|----------------------|-----------|
| tuit17        | .0453057    | .0224781  | 2.02  | 0.044 | .0012059             | .0894056  |
| 1.sib3        | -.7144889   | .3028995  | -2.36 | 0.018 | -1.308748            | -.1202302 |
| sib3#c.tuit17 |             |           |       |       |                      |           |
| 1             | -.0297482   | .0318676  | -0.93 | 0.351 | -.0922694            | .0327729  |
| _cons         | 13.16868    | .2116824  | 62.21 | 0.000 | 12.75338             | 13.58398  |

- Draw the relationship between years of education and college tuition at age 17 (*educ*) for those who have three or more siblings and those who have fewer. Label the slope and intercept terms. (Ignore the statistical significance of the estimates for now.)
- Can we conclude that the effect of college tuition at age 17 on years of education differs between those with three or more siblings and those with fewer? Explain.
- Redraw the relationship if necessary, based on your conclusion from (b).

**STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.**

| <b>Z</b>    | <b>.00</b> | <b>.01</b> | <b>.02</b> | <b>.03</b> | <b>.04</b> | <b>.05</b> | <b>.06</b> | <b>.07</b> | <b>.08</b> | <b>.09</b> |
|-------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| <b>-3.9</b> | .00005     | .00005     | .00004     | .00004     | .00004     | .00004     | .00004     | .00004     | .00003     | .00003     |
| <b>-3.8</b> | .00007     | .00007     | .00007     | .00006     | .00006     | .00006     | .00006     | .00005     | .00005     | .00005     |
| <b>-3.7</b> | .00011     | .00010     | .00010     | .00010     | .00009     | .00009     | .00008     | .00008     | .00008     | .00008     |
| <b>-3.6</b> | .00016     | .00015     | .00015     | .00014     | .00014     | .00013     | .00013     | .00012     | .00012     | .00011     |
| <b>-3.5</b> | .00023     | .00022     | .00022     | .00021     | .00020     | .00019     | .00019     | .00018     | .00017     | .00017     |
| <b>-3.4</b> | .00034     | .00032     | .00031     | .00030     | .00029     | .00028     | .00027     | .00026     | .00025     | .00024     |
| <b>-3.3</b> | .00048     | .00047     | .00045     | .00043     | .00042     | .00040     | .00039     | .00038     | .00036     | .00035     |
| <b>-3.2</b> | .00069     | .00066     | .00064     | .00062     | .00060     | .00058     | .00056     | .00054     | .00052     | .00050     |
| <b>-3.1</b> | .00097     | .00094     | .00090     | .00087     | .00084     | .00082     | .00079     | .00076     | .00074     | .00071     |
| <b>-3.0</b> | .00135     | .00131     | .00126     | .00122     | .00118     | .00114     | .00111     | .00107     | .00104     | .00100     |
| <b>-2.9</b> | .00187     | .00181     | .00175     | .00169     | .00164     | .00159     | .00154     | .00149     | .00144     | .00139     |
| <b>-2.8</b> | .00256     | .00248     | .00240     | .00233     | .00226     | .00219     | .00212     | .00205     | .00199     | .00193     |
| <b>-2.7</b> | .00347     | .00336     | .00326     | .00317     | .00307     | .00298     | .00289     | .00280     | .00272     | .00264     |
| <b>-2.6</b> | .00466     | .00453     | .00440     | .00427     | .00415     | .00402     | .00391     | .00379     | .00368     | .00357     |
| <b>-2.5</b> | .00621     | .00604     | .00587     | .00570     | .00554     | .00539     | .00523     | .00508     | .00494     | .00480     |
| <b>-2.4</b> | .00820     | .00798     | .00776     | .00755     | .00734     | .00714     | .00695     | .00676     | .00657     | .00639     |
| <b>-2.3</b> | .01072     | .01044     | .01017     | .00990     | .00964     | .00939     | .00914     | .00889     | .00866     | .00842     |
| <b>-2.2</b> | .01390     | .01355     | .01321     | .01287     | .01255     | .01222     | .01191     | .01160     | .01130     | .01101     |
| <b>-2.1</b> | .01786     | .01743     | .01700     | .01659     | .01618     | .01578     | .01539     | .01500     | .01463     | .01426     |
| <b>-2.0</b> | .02275     | .02222     | .02169     | .02118     | .02068     | .02018     | .01970     | .01923     | .01876     | .01831     |
| <b>-1.9</b> | .02872     | .02807     | .02743     | .02680     | .02619     | .02559     | .02500     | .02442     | .02385     | .02330     |
| <b>-1.8</b> | .03593     | .03515     | .03438     | .03362     | .03288     | .03216     | .03144     | .03074     | .03005     | .02938     |
| <b>-1.7</b> | .04457     | .04363     | .04272     | .04182     | .04093     | .04006     | .03920     | .03836     | .03754     | .03673     |
| <b>-1.6</b> | .05480     | .05370     | .05262     | .05155     | .05050     | .04947     | .04846     | .04746     | .04648     | .04551     |
| <b>-1.5</b> | .06681     | .06552     | .06426     | .06301     | .06178     | .06057     | .05938     | .05821     | .05705     | .05592     |
| <b>-1.4</b> | .08076     | .07927     | .07780     | .07636     | .07493     | .07353     | .07215     | .07078     | .06944     | .06811     |
| <b>-1.3</b> | .09680     | .09510     | .09342     | .09176     | .09012     | .08851     | .08691     | .08534     | .08379     | .08226     |
| <b>-1.2</b> | .11507     | .11314     | .11123     | .10935     | .10749     | .10565     | .10383     | .10204     | .10027     | .09853     |
| <b>-1.1</b> | .13567     | .13350     | .13136     | .12924     | .12714     | .12507     | .12302     | .12100     | .11900     | .11702     |
| <b>-1.0</b> | .15866     | .15625     | .15386     | .15151     | .14917     | .14686     | .14457     | .14231     | .14007     | .13786     |
| <b>-0.9</b> | .18406     | .18141     | .17879     | .17619     | .17361     | .17106     | .16853     | .16602     | .16354     | .16109     |
| <b>-0.8</b> | .21186     | .20897     | .20611     | .20327     | .20045     | .19766     | .19489     | .19215     | .18943     | .18673     |
| <b>-0.7</b> | .24196     | .23885     | .23576     | .23270     | .22965     | .22663     | .22363     | .22065     | .21770     | .21476     |
| <b>-0.6</b> | .27425     | .27093     | .26763     | .26435     | .26109     | .25785     | .25463     | .25143     | .24825     | .24510     |
| <b>-0.5</b> | .30854     | .30503     | .30153     | .29806     | .29460     | .29116     | .28774     | .28434     | .28096     | .27760     |
| <b>-0.4</b> | .34458     | .34090     | .33724     | .33360     | .32997     | .32636     | .32276     | .31918     | .31561     | .31207     |
| <b>-0.3</b> | .38209     | .37828     | .37448     | .37070     | .36693     | .36317     | .35942     | .35569     | .35197     | .34827     |
| <b>-0.2</b> | .42074     | .41683     | .41294     | .40905     | .40517     | .40129     | .39743     | .39358     | .38974     | .38591     |
| <b>-0.1</b> | .46017     | .45620     | .45224     | .44828     | .44433     | .44038     | .43644     | .43251     | .42858     | .42465     |
| <b>-0.0</b> | .50000     | .49601     | .49202     | .48803     | .48405     | .48006     | .47608     | .47210     | .46812     | .46414     |



★ Read the questions carefully  
 ★ Make sure that you answered everything

**Problem 1.** Use the regression results below to answer the questions.  $lprice$  log(price, \$1,000),  $lsqrtft$  log(size of house in square feet),  $llotsize$  log(size of lot in square feet), *colonial* colonial style (=1), *bdrms* number of bedrooms

. reg lprice lsqrtft llotsize colonial bdrms

| Source   | SS         | df | MS         | Number of obs | = | 88     |
|----------|------------|----|------------|---------------|---|--------|
| Model    | 5.20397919 | 4  | 1.3009948  | F(4, 83)      | = | 38.38  |
| Residual | 2.81362433 | 83 | .033899088 | Prob > F      | = | 0.0000 |
|          |            |    |            | R-squared     | = | 0.6491 |
|          |            |    |            | Adj R-squared | = | 0.6322 |
| Total    | 8.01760352 | 87 | .092156362 | Root MSE      | = | .18412 |

| <u>lprice</u>   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|-----------------|-------------|-----------|-------|-------|----------------------|-----------|
| lsqrtft         | .7071931    | .092802   | 7.62  | 0.000 | .5226138             | .8917725  |
| llotsize        | .1678189    | .0381807  | 4.40  | 0.000 | .0918791             | .2437587  |
| <u>colonial</u> | .0537962    | .0447732  | 1.20  | 0.233 | -.035256             | .1428483  |
| bdrms           | .0268305    | .0287236  | 0.93  | 0.353 | -.0302995            | .0839605  |
| _cons           | -1.349589   | .651041   | -2.07 | 0.041 | -2.644483            | -.0546947 |

(a) Interpret the coefficient on  $lsqrtft$

log-log      A 1% increase in the size of the house increases the housing price by 0.71%, all else equal.

(b) Interpret the coefficient on *colonial*

log-lin      Housing prices are 5.38% higher for colonial-style houses compared to non-colonial style houses, all else equal. However, the estimate is not statistically sig. at the 5% lv.

(c) Interpret the coefficient on *bdrms*

log-lin      An add'l bedroom increases housing prices by 2.68%, all else equal. However, the estimate is not statistically sig. at the 5% lv.

**Problem 2.** Based on the table below, answer the questions.

| Dependent Variable=Labor Force Participation |                                       |                    |                    | ↓                  | ↓                  |
|----------------------------------------------|---------------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                              |                                       | (1)                | (2)                | (3)                | (4)                |
|                                              |                                       | LPM                | Probit             | LPM                | Probit             |
| $\beta_1$                                    | HHI                                   | -0.11**<br>(0.006) | -0.38**<br>(0.019) | -0.09**<br>(0.006) | -0.32**<br>(0.021) |
| $\beta_2$                                    | LF experience                         |                    |                    | -0.01**<br>(0.000) | -0.03**<br>(0.001) |
| $\beta_3$                                    | nchild<6yrs old                       |                    |                    | -0.14**<br>(0.005) | -0.49**<br>(0.016) |
| $\beta_4$                                    | nchild 6~18yrs old                    |                    |                    | -0.02**<br>(0.003) | -0.09**<br>(0.010) |
| $\beta_5$                                    | Husband's earned income<br>(\$10,000) |                    |                    | -0.01**<br>(0.001) | -0.02**<br>(0.004) |
| $\beta_0$                                    | Constant                              | 0.83**<br>(0.004)  | 0.97**<br>(0.016)  | 1.09**<br>(0.008)  | 1.97**<br>(0.039)  |
|                                              | Observations                          | 22,369             | 22,369             | 22,259             | 22,259             |
|                                              | R-squared                             | 0.017              |                    | 0.071              |                    |

Robust standard errors in parentheses  
\*\* p<0.01, \* p<0.05, + p<0.1

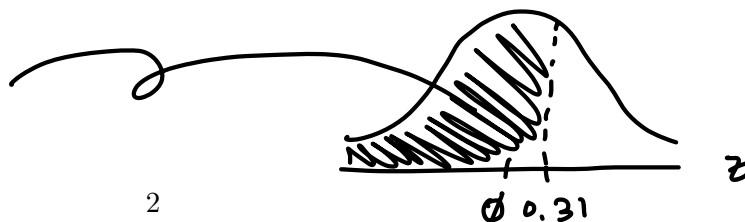
$$P(LFP=1) = \beta_0 + \beta_1 HHI + \beta_2 exp + \beta_3 nch6 + \beta_4 nch18 + \beta_5 hinc + u$$

- (a) Suppose a married woman is covered by her husband's insurance (HHI), has 10 years of working experience, and 2 children who are younger than 6 years old and her husband's income is \$30,000. In column (3), what is the probability that the woman will participate in the labor force?

$$\begin{aligned}
 P(LFP=1 \mid HHI=1, exp=10, nchild<6=2, nchild6-18=0, hinc=3) \\
 &= 1.09 + (-0.09) + (-0.01) \times 10 + (-0.14) \cdot 2 + 0 + (-0.01) \cdot 3 \\
 &= 0.59
 \end{aligned}$$

- (b) In column (4), what is the probability that the woman will participate in the labor force?

$$\begin{aligned}
 P(LFP=1 \mid HHI=1, exp=10, nchild<6=2, nchild6-18=0, hinc=3) \\
 &= \Phi(1.97 - 0.32 + (-0.03) \cdot 10 + (-0.49) \cdot 2 + 0 + (-0.02) \cdot 3) \\
 &= \Phi(0.31) \\
 &= 0.6217
 \end{aligned}$$



→ I added this question to help you review Q2 in Discussion 3.

**Problem 3.** Using data from a private school voucher program, also known as a school choice program, we estimated the following regression. *choicelyrs* represents the number of years a student attended a choice school from 1991 to 1994. Since it is a categorical variable that takes the values 0, 1, 2, 3, and 4, we generated binary variables *choicelyrs1*, *choicelyrs2*, *choicelyrs3*, and *choicelyrs4*.

```
. reg mathscore choicelyrs1 choicelyrs2 choicelyrs3 choicelyrs4 black hispanic female
```

| Source   | SS         | df  | MS         | Number of obs | = | 990    |
|----------|------------|-----|------------|---------------|---|--------|
| Model    | 37872.1524 | 7   | 5410.30749 | F(7, 982)     | = | 13.52  |
| Residual | 392958.98  | 982 | 400.161894 | Prob > F      | = | 0.0000 |
|          |            |     |            | R-squared     | = | 0.0879 |
|          |            |     |            | Adj R-squared | = | 0.0814 |
| Total    | 430831.132 | 989 | 435.622985 | Root MSE      | = | 20.004 |

| mathscore   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|-------------|-------------|-----------|-------|-------|----------------------|-----------|
| choicelyrs1 | .5102603    | 1.735531  | 0.29  | 0.769 | -2.895516            | 3.916037  |
| choicelyrs2 | -1.314817   | 2.098575  | -0.63 | 0.531 | -5.433025            | 2.803392  |
| choicelyrs3 | -2.766788   | 2.198078  | -1.26 | 0.208 | -7.080257            | 1.546681  |
| choicelyrs4 | -.617641    | 2.856311  | -0.22 | 0.829 | -6.222817            | 4.987535  |
| black       | -16.19045   | 1.849133  | -8.76 | 0.000 | -19.81916            | -12.56174 |
| hispanic    | -13.37447   | 2.40842   | -5.55 | 0.000 | -18.10072            | -8.648231 |
| female      | 1.370047    | 1.277174  | 1.07  | 0.284 | -1.136258            | 3.876351  |
| _cons       | 57.04585    | 1.660486  | 34.35 | 0.000 | 53.78734             | 60.30436  |

- (a) Interpret the coefficient on *choicelyrs1* (omitted baseline = *choicelyrs* 0)

Students who attended a choice school for one year

scored 0.51 points higher in math than those who never attended, all else equal.

- (b) How would you rewrite a regression model to examine whether math scores differ between students who have ever attended a choice school and those who have never attended?

baseline

① We should generate a new dummy variable which takes the value of 1 if a student attended a choice school, and 0 otherwise.

→  $choice = \begin{cases} 1 & \text{if a student attended a choice school} \\ 0 & \text{have never attended} \end{cases}$

② We should run the following model:

$$mathscore = \beta_0 + \beta_1 choice + \beta_2 black + \beta_3 hispanic + \beta_4 female + u$$

difference in math score btw students who attended a choice school & tho who did not.

**Problem 4.** The following regression results are based on a dataset of working men from 1991, with *abil* representing cognitive ability and *educ* representing years of education.

. reg educ motheduc fatheduc abil abil2

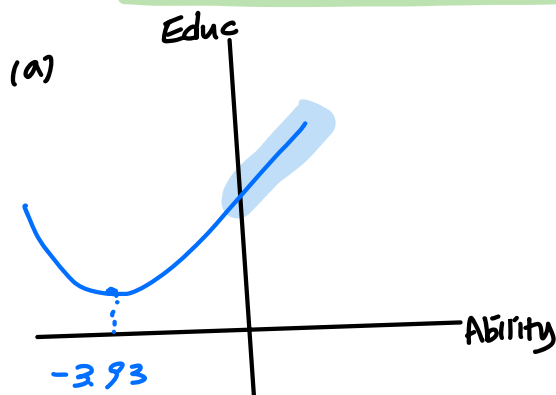
| Source   | SS         | df    | MS         | Number of obs | = | 1,230  |
|----------|------------|-------|------------|---------------|---|--------|
|          |            |       |            | F(4, 1225)    | = | 244.91 |
| Model    | 3027.03706 | 4     | 756.759264 | Prob > F      | = | 0.0000 |
| Residual | 3785.24262 | 1,225 | 3.08999397 | R-squared     | = | 0.4444 |
|          |            |       |            | Adj R-squared | = | 0.4425 |
| Total    | 6812.27967 | 1,229 | 5.54294522 | Root MSE      | = | 1.7578 |

| educ     | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |          |
|----------|-------------|-----------|-------|-------|----------------------|----------|
| motheduc | .1901261    | .0280957  | 6.77  | 0.000 | .1350051             | .2452472 |
| fatheduc | .1089387    | .0196014  | 5.56  | 0.000 | .0704827             | .1473946 |
| abil     | .4014624    | .0302875  | 13.26 | 0.000 | .3420413             | .4608835 |
| abil2    | .050599     | .0083039  | 6.09  | 0.000 | .0343076             | .0668905 |
| _cons    | 8.240226    | .2874099  | 28.67 | 0.000 | 7.676356             | 8.804097 |

$$\text{Educ} = \beta_0 + \beta_1 \text{abil} + \beta_2 \text{abil}^2 + \beta_3 \text{moedu} + \beta_4 \text{fatheduc} + u$$

- (a) Draw the relationship between cognitive ability (*abil*) and years of education (*educ*).
- (b) Find the turning point.
- (c) Conduct a significance test at the 5% level to evaluate whether the relationship between cognitive ability and years of education is linear.



(b) Turning pt: slope = 0

$$\frac{\partial \text{Educ}}{\partial \text{Abil}} = 0$$

$$\Rightarrow \beta_1 + 2\beta_2 \text{Abil} = 0$$

$$\text{Abil}^* = -\frac{\beta_1}{2\beta_2} = -\frac{0.401}{2(0.05)} = -3.93$$

not meaningful  
b/c ability is not negative  
(can't be)

(c) ①  $H_0: \beta_2 = 0$   
 $H_a: \beta_2 \neq 0$

② sig. lv  $\alpha = 0.05$

③ test stat =  $\frac{\hat{\beta}_2 - \beta_{2,0}}{\text{se}(\hat{\beta}_2)} = \frac{\hat{\beta}_2}{\text{se}(\hat{\beta}_2)}$   
 $= \frac{0.050599}{0.0083039} = 6.09$

④ p-value =  $P(|t| > |\text{test-stat}|)$

$$= P(|t| > 6.09)$$

$$\approx 2 \times P(Z < -6.09) = 2 \times \phi = \phi$$

⑤ Conclusion

p-value of  $\phi$  is smaller than the sig. lv 0.05.

Therefore, we reject  $H_0$  that  $\beta_2 = 0$   
 $\therefore$  Educ & Ability have a nonlinear rel.

**Problem 5.** Using the same dataset, we run a different regression to examine the effect of number of siblings and college tuition. *sibs* is the number of siblings and *tuit17* is college tuition at age 17.

→ gen sib3 = *sibs* >= 3      ← dummy       $sib3 = \begin{cases} 1 & \text{if } \#sibs \geq 3 \\ 0 & \text{o/w} \end{cases}$

→ reg educ c.tuit17##sib3 = *tuit17 sib3*      *tuit17 x sib3*

|          |            |       |            |               |   |        |
|----------|------------|-------|------------|---------------|---|--------|
| Source   | SS         | df    | MS         | Number of obs | = | 1,230  |
|          |            |       |            | F(3, 1226)    | = | 19.19  |
| Model    | 305.529838 | 3     | 101.843279 | Prob > F      | = | 0.0000 |
| Residual | 6506.74984 | 1,226 | 5.30730003 | R-squared     | = | 0.0448 |
|          |            |       |            | Adj R-squared | = | 0.0425 |
| Total    | 6812.27967 | 1,229 | 5.54294522 | Root MSE      | = | 2.3038 |

|           | educ          | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |
|-----------|---------------|-------------|-----------|-------|-------|----------------------|
| $\beta_1$ | tuit17        | .0453057    | .0224781  | 2.02  | 0.044 | .0012059 .0894056    |
| $\beta_2$ | 1.sib3        | -.7144889   | .3028995  | -2.36 | 0.018 | -1.308748 -.1202302  |
| $\beta_3$ | sib3#c.tuit17 | -.0297482   | .0318676  | -0.93 | 0.351 | -.0922694 .0327729   |
| $\beta_0$ | _cons         | 13.16868    | .2116824  | 62.21 | 0.000 | 12.75338 13.58398    |

$$Educ = \beta_0 + \beta_1 \text{tuit17} + \beta_2 \text{sib3} + \beta_3 (\text{tuit17} \times \text{sib3}) + u$$

- (a) Draw the relationship between years of education and college tuition at age 17 (*educ*) for those who have ~~more than three~~ <sup>three or more</sup> siblings and those who have fewer than three. Label the slope and intercept terms. (Ignore the statistical significance of the estimates for now.)

- (b) Can we conclude that the effect of college tuition at age 17 on years of education differs between those with more ~~than three~~ <sup>three or</sup> siblings and those with ~~three or fewer~~? Explain.  $\Rightarrow \beta_3$

- (c) Redraw the relationship if necessary, based on your conclusion from (b).

No, the estimated coeff on the interaction term is not statistically sig. at the 5% lv.

**3+ siblings**

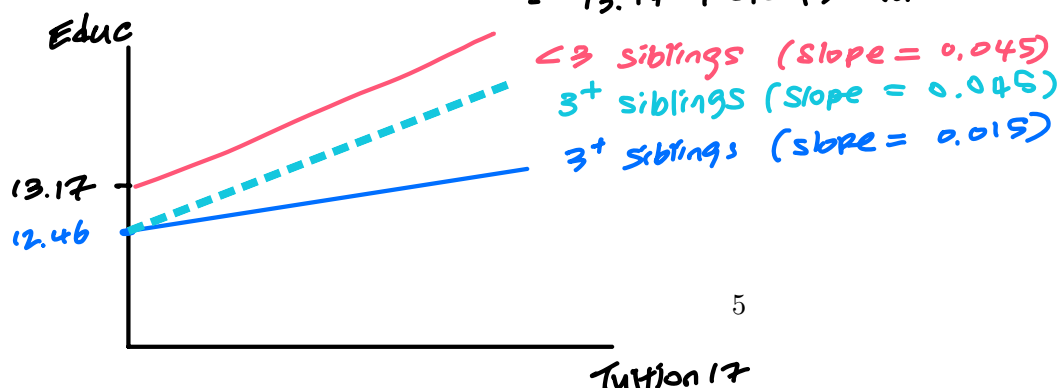
$$E[educ | \text{tuit17}, \text{sib3}=1] = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot \text{tuit17}$$

$$= 12.46 + 0.015 \text{tuit17} \quad \checkmark$$

**<3 siblings**

$$E[educ | \text{tuit17}, \text{sib3}=0] = \beta_0 + \beta_1 \text{tuit17}$$

$$= 13.17 + 0.045 \text{tuit17}$$



$\Rightarrow$  3+ sib group and <3 sib group have the same slope.

**Problem 1.** You decided to examine how macroeconomic conditions affect drug overdose deaths. Suppose you ran the following simple regression using county-level data and obtained the results. The dependent variable, Overdose Death Rate, represents the number of deaths per 100,000 people, and the variable of interest, Unemployment Rate, is measured in percentage.

$$\text{Overdose Death Rate}_i = \beta_0 + \beta_1 \text{Unemployment Rate}_i + u_i$$

- (a) Interpret the coefficients. (Assume that  $\beta_0, \beta_1 > 0$ .)
  
  
  
  
  
  
  
  
  
  
- (b) Do you think the regression suffers from omitted variable bias? If so, give one example of a confounder and predict the direction of the omitted variable bias due to the confounder.
  
  
  
  
  
  
  
  
  
  
- (c) Hollingsworth et al. (2017) use a panel data set and run the same regression model but with county and year fixed effects. According to their result, the estimated coefficient of the Unemployment Rate is (0.22) with the standard error of 0.05. Interpret the result.

**Problem 2.** Currie and Walker (2011) examine the impact of introducing E-ZPass, an electronic toll collection system that reduced traffic congestion at toll plazas and, in turn, lowered pollution levels. They assess how this change affected infant health by comparing mothers living near toll plazas. In their DD design, the treatment group consists of mothers residing within 2 km of a toll plaza, while the control group includes those living between 2 km and 10 km from a toll plaza but still near a highway.

$$Y_{it} = \beta_0 + \beta_1 \text{E-ZPass}_{it} + \beta_2 \text{Close}_{it} + \beta_3 (\text{E-ZPass}_{it} \times \text{Close}_{it}) + u_{it}$$

E-ZPass is an indicator equal to one if the closest toll plaza has implemented E-ZPass; close it is an indicator equal to one if the mother lived within 2 km of a toll plaza.

Fill in the table in terms of  $\beta_0, \beta_1, \beta_2$ , and  $\beta_3$

|                           | <2 km E-ZPass | >2 km and <10 km E-ZPass |
|---------------------------|---------------|--------------------------|
| Low birth weight (before) |               |                          |
| Low birth weight (after)  |               |                          |

The summary statistics show the following:

|                                 | <2 km E-ZPass | >2 km and <10 km E-ZPass |
|---------------------------------|---------------|--------------------------|
| Low birth weight (before)       | 0.082         | 0.089                    |
| Low birth weight (after)        | 0.078         | 0.092                    |
| <i>Difference, after-before</i> |               |                          |

Source: Currie and Walker (2011) Table 1.

(a) Find the causal effect of E-ZPass on low birth weight.

(b) In the space below, draw the following:

- Low birth weight before and after for mothers living within 2 km of an E-ZPass.
- Low birth weight before and after for mothers living between 2 km and 10 km of an E-ZPass.
- Low birth weight for mothers living within 2 km of an E-ZPass in the absence of E-ZPass adoption.
- The treatment effect.

**Problem 3.** In the U.S., individuals become eligible for Medicare when they turn 65. Many economists have used this age threshold to study the effect of health insurance on the utilization of health care services. Suppose we have monthly insurance claims data for individuals from 12 months before to 12 months after their 65th birthday.

(a) Fill out the following:

- Treatment:
- Outcome:
- Running variable:
- Cutoff:
- Treated individuals:
- Control individuals:

(b) Write out an equation to estimate an RD design with different slopes before and after the cutoff. Draw an RD graph assuming that all the coefficients are positive. Label the following:

- The treatment and control groups, the x- and y-axes, and the cutoff
- $E[Y|\text{Treat} = 1]$
- $E[Y|\text{Treat} = 0]$
- $E[Y|\text{Treat} = 1]$  at the cutoff
- $E[Y|\text{Treat} = 0]$  at the cutoff
- The treatment effect

**Problem 1.** You decided to examine how macroeconomic conditions affect drug overdose deaths. Suppose you ran the following simple regression using county-level data and obtained the results. The dependent variable, Overdose Death Rate, represents the number of deaths per 100,000 people, and the variable of interest, Unemployment Rate, is measured in percentage.

$$\text{Overdose Death Rate}_i = \beta_0 + \beta_1 \text{Unemployment Rate}_i + u_i$$

- (a) Interpret the coefficients. (Assume that  $\beta_0, \beta_1 > 0$ .)

$\beta_0$  : Expected odd rate when ue rate is zero.

$\beta_1$  : if ue rate increases by 1pp, the expected odd rate increases by  $\beta_1$ .

- (b) Do you think the regression suffers from omitted variable bias? If so, give one example of a confounder and predict the direction of the omitted variable bias due to the confounder.

Poverty rate

True : odd rate =  $\beta_0 + \beta_1 \text{UE rate} + \beta_2 \text{Poverty rate} + u_i$

Aux : Pov rate =  $\delta_0 + \delta_1 \text{UE rate}$

Bias =  $\beta_2 \cdot \delta_1 > 0$   $\leftarrow$  explain  $\Rightarrow$  An upward bias

- (c) Hollingsworth et al. (2017) use a panel data set and run the same regression model but with county and year fixed effects. According to their result, the estimated coefficient of the Unemployment Rate is (0.22) with the standard error of 0.05. Interpret the result.

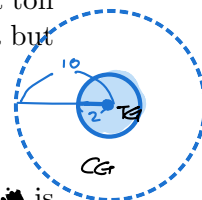
if a county's UE rate increases by 1pp.

the odd rate rises by 0.22, controlling for the year FEs.

**Problem 2.** Currie and Walker (2011) examine the impact of introducing E-ZPass, an electronic toll collection system that reduced traffic congestion at toll plazas and, in turn, lowered pollution levels. They assess how this change affected infant health by comparing mothers living near toll plazas. In their DD design, the treatment group consists of mothers residing within 2 km of a toll plaza, while the control group includes those living between 2 km and 10 km from a toll plaza but still near a highway.

$$Y_{it} = \beta_0 + \beta_1 \text{E-ZPass}_{it} + \beta_2 \text{Close}_{it} + \beta_3 (\text{E-ZPass}_{it} \times \text{Close}_{it}) + u_{it}$$

E-ZPass is an indicator equal to one if the closest toll plaza has implemented E-ZPass; close is an indicator equal to one if the mother lived within 2 km of a toll plaza.



Fill in the table in terms of  $\beta_0, \beta_1, \beta_2$ , and  $\beta_3$

$EZ = \emptyset$

$EZ = 1$

|                           | $close = 1$<br><2 km E-ZPass                             | $close = \emptyset$<br>>2 km and <10 km E-ZPass |
|---------------------------|----------------------------------------------------------|-------------------------------------------------|
| Low birth weight (before) | $E(Y c=1, EZ=\emptyset) = \beta_0 + \beta_2$             | $E(Y c=\emptyset, EZ=\emptyset) = \beta_0$      |
| Low birth weight (after)  | $E(Y c=1, EZ=1) = \beta_0 + \beta_1 + \beta_2 + \beta_3$ | $E(Y c=\emptyset, EZ=1) = \beta_0 + \beta_1$    |

The summary statistics show the following:

|                           | <2 km E-ZPass            | >2 km and <10 km E-ZPass |
|---------------------------|--------------------------|--------------------------|
| Low birth weight (before) | 0.082                    | 0.089                    |
| Low birth weight (after)  | 0.078                    | 0.092                    |
| Difference, after-before  | $0.078 - 0.082 = -0.004$ | $0.092 - 0.089 = 0.003$  |

$\Rightarrow \beta_{DD}$

Source: Currie and Walker (2011) Table 1.

$= -0.004$   
 $- 0.003$

(a) Find the causal effect of E-ZPass on low birth weight.

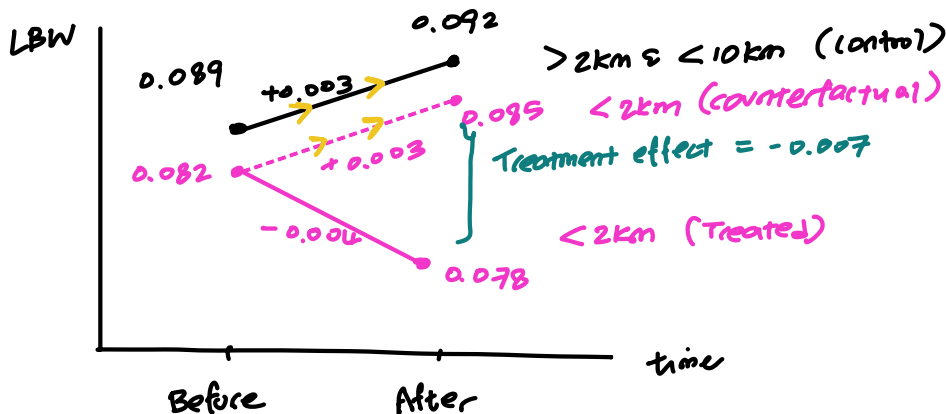
$$E(Y|c=1, EZ=1) - E(Y|c=1, EZ=\emptyset) = -0.004$$

$$E(Y|c=\emptyset, EZ=1) - E(Y|c=\emptyset, EZ=\emptyset) = 0.003$$

$$\therefore \text{Causal effect} = \Delta_{<2km} - \Delta_{>2km \& <10km} = -0.004 - 0.003 = -0.007$$

(b) In the space below, draw the following:

- Low birth weight before and after for mothers living within 2 km of an E-ZPass.
- Low birth weight before and after for mothers living between 2 km and 10 km of an E-ZPass.
- Low birth weight for mothers living within 2 km of an E-ZPass in the absence of E-ZPass adoption.
- The treatment effect.



**Problem 3.** In the U.S., individuals become eligible for Medicare when they turn 65. Many economists have used this age threshold to study the effect of health insurance on the utilization of health care services. Suppose we have monthly insurance claims data for individuals from 12 months before to 12 months after their 65th birthday.

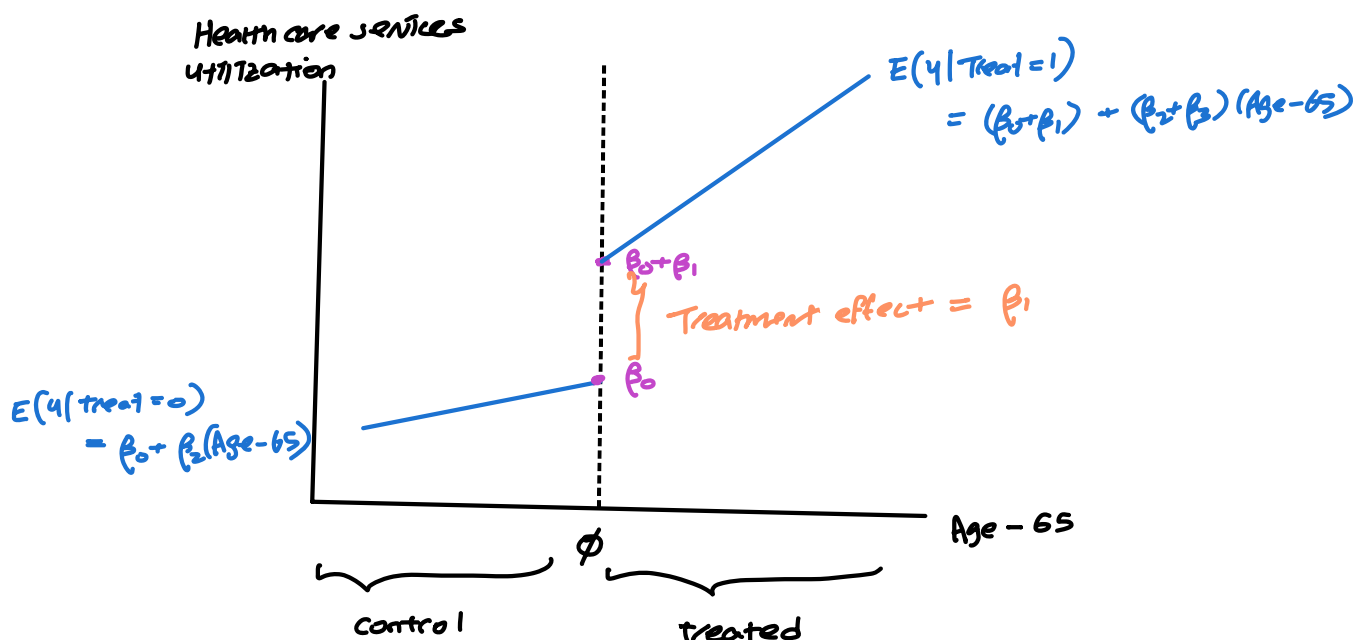
(a) Fill out the following:

- Treatment: *being eligible for Medicare*
- Outcome: *use of healthcare services*
- Running variable: *Age*
- Cutoff: *65*
- Treated individuals: *Age  $\geq 65$*
- Control individuals: *Age  $< 65$*

(b) Write out an equation to estimate an RD design with different slopes before and after the cutoff. Draw an RD graph assuming that all the coefficients are positive. Label the following:

- The treatment and control groups, the x- and y-axes, and the cutoff
- $E[Y|\text{Treat} = 1] = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)(\text{Age} - 65)$
- $E[Y|\text{Treat} = 0] = \beta_0 + \beta_2(\text{Age} - 65)$
- $E[Y|\text{Treat} = 1]$  at the cutoff  $\beta_0 + \beta_1$
- $E[Y|\text{Treat} = 0]$  at the cutoff  $\beta_0$
- The treatment effect  $\beta_1$

$$y_i = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \overbrace{(\text{Age} - 65)}^{\text{distance}} + \beta_3 \overbrace{[\text{Treat} \times (\text{Age} - 65)]}^{\text{distance}} + u_i$$



## Binary Dependent Variable Models

### I. Linear Probability Model (LPM)

A **linear** regression model that  $y$  is a binary dependent variable.

---


$$y_i = E[y|x] + u_i \quad \text{Since } E[y|x] = Pr[y = 1|x] \times 1 + Pr[y = 0|x] \times 0, \quad (1)$$

$$= Pr[y = 1|x] + u_i \quad E[y|x] = Pr[y = 1|x] \quad (2)$$

$$= \beta_0 + \beta_1 x_i + u_i \quad (3)$$


---

### Interpretation

Change in  $x \Rightarrow$  Change in the probability that  $y=1$

- $\widehat{\beta}_0$  : When  $x=0$ , the probability that  $y=1$  is  $\widehat{\beta}_0$ , on average.
- $\widehat{\beta}_1$  : An increase in  **$x$**  by one-unit is associated with an increase/decrease in **the probability that  $y=1$**  by  $\widehat{\beta}_1$ , on average.

## II. **Probit model**

We use **standard normal distribution (=Z-distribution)** to model the probability that  $y=1$ .

$$y_i = E[y|x] + u_i \quad E[y|x] = Pr[y = 1|x] \times 1 + Pr[y = 0|x] \times 0 = Pr[y = 1|x] \quad (1)$$

$$= Pr[y = 1|x] + u_i \quad (2)$$

$$= \Phi(\beta_0 + \beta_1 x) + u_i \quad , \Phi(.) \text{ the cumulative standard normal distribution function} \quad (3')$$

$$= \Phi(z) + u_i \quad , \text{ where } z = \beta_0 + \beta_1 x, \text{ and } z \sim N(0,1) \quad (4')$$

y: Binary dependent variable

### Interpretation

- $\widehat{\beta}_0$  : When  $x=0$ , the probability of  $y=1$  is  $\Phi(\widehat{\beta}_0)$
- $\widehat{\beta}_1$  : A one unit increase in **x** changes the **z-score** by  $\widehat{\beta}_1$  on average.
  - Remember  $z = \beta_0 + \beta_1 x$ ! (i.e.  $\widehat{\beta}_1$  is the relationship between  $x$  and  $z$ -score!)

### A. **Finding the predicted probability that $y=1$**

(i.e.) Finding  $\widehat{y} = E[\widehat{y}|x] = Pr(\widehat{y} = 1|x) = \Phi(\widehat{\beta}_0 + \widehat{\beta}_1 x)$

- 1) Calculate the Z-score= $\widehat{\beta}_0 + \widehat{\beta}_1 x$
- 2) Look up the Z-score in the standard normal distribution table.  $\Phi(\widehat{\beta}_0 + \widehat{\beta}_1 x)$

### B. **The effect of a change in x on the change in the predicted probability ( $\widehat{y}$ ).**

*Again, be careful! The effect of a change in x on the predicted probability is NOT  $\widehat{\beta}_1$*

- 1) Compute the predicted probability for the initial value of x, let's say  $x_0$ . (Follow the steps in A)
- 2) Compute the predicted probability for the changed value of x, let's say  $x_1 = x_0 + \Delta x$
- 3) Take the difference between 1) and 2).

$\Rightarrow$  When  $x$  changes by  $\Delta x$ , from  $x_0$  to  $x_1$ , the predicted probability,  $\widehat{y}$ , changes by

$$\Phi(\widehat{\beta}_0 + \widehat{\beta}_1 x_1) - \Phi(\widehat{\beta}_0 + \widehat{\beta}_1 x_0).$$

$$\therefore Pr(\widehat{y} = 1|x_1) - Pr(\widehat{y} = 1|x_0) = \Phi(\widehat{\beta}_0 + \widehat{\beta}_1 x_1) - \Phi(\widehat{\beta}_0 + \widehat{\beta}_1 x_0).$$

### III. **Logit model**

We use **cumulative logistic function** to model the probability that  $y=1$ .

---


$$y_i = E[y|x] + u_i \quad E[y|x] = Pr[y = 1|x] \times 1 + Pr[y = 0|x] \times 0 = Pr[y = 1|x] \quad (1)$$

$$= Pr[y = 1|x] + u_i \quad (2)$$

$$= F(\beta_0 + \beta_1 x) + u_i \quad , \text{ where } F(\alpha) = \frac{1}{1+e^{-\alpha}} \text{ is a cumulative logistic function} \quad (3'')$$

$$= \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} + u_i \quad (4'')$$


---

$y$ : Binary dependent variable

#### Interpretation

- $\widehat{\beta}_0$  : The log odds of  $y=1$  when  $x=0$ .
- $\widehat{\beta}_1$  : A one unit increase in  $x$  changes **the log odds of  $y=1$**  by  $\widehat{\beta}_1$ , on average<sup>1</sup>.

---


$$^1 \text{ odds} = \frac{\text{probability that } y=1}{\text{probability that } y=0} = \frac{\text{probability that } y=1}{(1-\text{probability that } y=1)}$$

$$\log \text{ odds} = \log (\text{odds}) = \text{logistic function}$$

Suppose  $Y=1$  is getting a job offer, and  $X$  is GPA.

Interpretation of Logit: A one point increase in GPA increases/decreases **the log odds of getting a job offer** by  $\widehat{\beta}_1$ , on average.

$$* \text{log odds of getting a job offer} = \log \left( \frac{\text{probability that you get a job offer}}{\text{probability that you don't get a job offer}} \right)$$

\*\* Statistically, we use log odds instead of odds because the shape of log odds (=logistic function) is similar to the normal distribution. It is why logit and probit functions are similar to each other!

## ECON 108 Lab Section

### Week 1: Introduction to Stata

#### Agenda for week 1

- Play with Stata
- Practice with a problem set

*\*\* Use help command, Google, and YouTube. They know everything. \*\**

#### I. Play with Stata

##### 1. Warming up and Opening a data set

1) \*: use it to indicate a comment. Lines beginning with a \* are ignored.

2) clear: clear the data space

3) set more off

: Stata pauses when there are too many lines to display. To continue, you should press any key.  
“set more off” command suppresses pauses and allow Stata to run smoothly.

4) cap log close

: capture the log close (Just in case you forget to close the log and run the do-file again.)

5) log using FILE\_NAME, replace

6) cd “PATH”: change directory

7) use FILE\_NAME, clear: import the file name

8) If your file is saved in a different folder other than your current working directory, you should load the file using the following command: use “[FILE\\_NAME\\_WITH\\_PATH](#)”  
You can obtain [FILE\\_NAME\\_WITH\\_PATH](#) by right clicking the data file.

9) help

: When you need help, type <help COMMAND>. STATA will give you the syntax, examples, and sometimes even mathematical proofs about the command. e.g. help reg

Example (do-file header)

```
*=====*
* ECON108 Lab Week1 *
* 01/08/2025 *
* my name *
*=====*

clear
set more off
cap log close
quietly log using ECON108_Lab_practice, replace

cd "/Users/Desktop/ECON108/DATA"
use FILENAME ,clear
```

## 2. Handling the data set

- 1) summarize (sum, su)
- 2) describe (des, d)
- 3) codebook
- 4) tab VARIABLE: tabulate
- 5) browse (bro): Browse the dataset
- 6) preserve/restore: Preserving the data for a while/Restoring the preserved data.
- 7) save DATANAME, [replace](#): save the current data as a dta file

*(\*You should include [replace](#) option to replace an old one to a new one.)\**

- 8) keep VAR1 VAR2 VAR5 (Keep only the listed variables)
- 9) drop VAR2 VAR4 (Drop all the listed variables)
- 10) sort: Arranging the observations into ascending order.

### 3. Playing with variables

1) generate(gen)

gen VARNAME=.

2) replace

replace VARNAME=1

replace VARNAME=1 if *condition*

3) label (lab)

label var VARNAME "This is VARNAME"

4) rename

rename OLDNAME NEWNAME

<Extra> You can rename all variables with lowercase

rename \*, lower

It means "Rename (rename) all variables (represented as \*) with lowercase option (, lower)."

5) egen: generating a function (mean, min, max, median and etc.)

egen NEWVAR = [function](#)(EXISTING\_VAR)

egen *mean\_age* = mean(age)

It means "Generate a variable named *mean\_age* as the average of the *age* variable across all observations."

help egen ([\\*Check the help file for available functions.](#))

### Example

Suppose you want to create a female dummy using an existing gender variable which takes the value of 1 if gender is male, and 2 if it is female.

- *Method1*

```
gen female=0
replace female=1 if gender==2
```

- *Method2*

```
gen female=gender==2
```

- *Method3*

```
gen female=.
replace female=0 if gender==1
replace female=1 if gender==2
lab var female "Female dummy"
```

*Wait! What is the difference between = and ==? Ask Google! ([Answer](#))*

*See? Google knows everything. Take advantage of all the resources out there. That will help you learn STATA fast.*

## **4. Graphs**

- 1) Histogram(hist)

```
hist VARNAME
```

```
help hist
```

- 2) Plotting Twoway graphs (scatter, line, and etc.)

```
twoway scatter Y X           *OR scatter Y X
```

```
twoway line Y X           *OR line Y X
```

```
help twoway
```

- 3) Saving a graph

```
graph export NAME.format, replace
```

```
help graph export (\*Check the help file for available format options.)
```

## 5. OLS

reg Y X

reg Y X, robust \*regression with robust standard errors

help reg

## 6. Producing a regression table with outreg2

outreg2 using FILENAME, replace

outreg2 using FNAME, excel replace

\*OR outreg2 using FNMAE.xls, replace

outreg2 using FNAME, word replace

\*OR outreg2 using FNAME.doc, replace

help outreg2

ssc install outreg2

(\*You need to install the package usually once, the very first time you use the command.)

### Producing a table with multiple regression results

reg lwage age edu female exper \*This is model (1)\*

outreg2 using table1, replace

reg lwage age age\_sq edu female exper \*This is model (2)\*

outreg2 using table1

reg lwage age age\_sq edu female exper nchild \*This is model (3)\*

outreg2 using table1

\*\*Include *replace* option only after the first model. Otherwise, the next model will replace the existing one.

## II. Practice

Please prepare your answers using a word processing program of your choice and then upload your answers to Canvas in either PDF or DOC format. For this lab you will be using College Scorecard data (DataScoreCard.dta). **Each observation** represents **a university** in the U.S.

Variables of Interest:

| Variable Name   | Description                                                      |
|-----------------|------------------------------------------------------------------|
| INSTNM          | Name of institution                                              |
| CITY            | City of institution                                              |
| AVGFACSL        | Average salary paid to faculty at the institution                |
| MEANEARNING6YRS | Average earnings six years after graduation from the institution |

**Always:**

1. Write down the relevant Stata command you used to answer a question.
  2. Show the relevant Stata output you used to answer the question.
  3. Write your interpretation of the output
- 
1. Suppose you wanted to investigate the relationship between postgraduate earnings and faculty salaries. Discuss whether you would expect a positive, negative, or no relationship between these two variables. [\[Write-up\]](#)
  2. How many universities are in this data set? What is the average postgraduate earnings 6 years after graduation for universities? How many observations have non-missing postgraduate earnings? What is the average salary paid to faculty? How many observations have non-missing faculty salary?
  3. Use the browse command to find University of California-Riverside (browse if CITY=="Riverside"). What are the mean earnings for graduates 6years after leaving UCR?
  4. Use the scatter command to plot the relationship between earnings 6 years after graduation and average faculty salary. What does this suggest about the relationship between the two variables?
  5. Run a regression of postgraduate earnings (dependent variable) on faculty salaries (explanatory variable). Why did we choose postgraduate earnings as the dependent variable? Interpret the slope coefficient. Interpret the intercept coefficient. Is the intercept coefficient economically meaningful?

6. Refer to your output in question 5. How many observations were used to estimate your regression? Why is this number less than the number of observations in the dataset? [\[Write-up\]](#)
7. Use the browse command to find universities in Riverside or Irvine. Which universities were not used in your regression in (5)? How can you tell?
8. What would be the expected increase in postgraduate earnings if UCR decided to increase faculty salaries by \$10,000? What would be the expected increase in postgraduate earnings if UCI decided to increase faculty salaries by \$10,000?
9. Using the predict command, obtain a predicted value for each university. Why does the predicted value for a university differ from its actual value?
10. Use the browse command to find UCR. Did the regression model overestimate or underestimate earnings?
11. If we knew the true value of the beta parameters, would we be able to perfectly predict each person's wages? Explain. [\[Write-up\]](#)
12. Use the gen command to generate a residual (that is the difference between the actual and the predicted value) for each university. How is the residual different from the error term?
13. Use the browse command to find UCR. What is the value of residual? Provide an interpretation.
14. Predict the expected postgraduate earnings for a student who attends a university that pays its faculty \$80,000. [\[Write-up\]](#)
15. Run a regression of postgraduate earnings on a variable of your choice. Explain the reasoning behind your choice. Interpret the results. Are the results what you would have expected? Why?

## ECON 108 Lab Section

### Week 2: Interpreting Stata output

#### Agenda for week 2

- Take a closer look at the estimation results in Stata
- Start Lab Assignment 1

Last week, we analyzed whether average faculty salaries influence students' postgraduate earnings using college-level data.

$$\text{Postgraduate earnings}_i = \beta_0 + \beta_1 \text{Faculty Salary}_i + u_i$$

Corresponding Stata command

`reg y x`

`reg MEANEARNING6YRS AVGFACSAL`

`. reg MEANEARNING6YRS AVGFACSAL`

| Source   | SS         | df    | MS         | Number of obs | = | 2,922   |
|----------|------------|-------|------------|---------------|---|---------|
| Model    | 1.4931e+11 | 1     | 1.4931e+11 | F(1, 2920)    | = | 1806.16 |
| Residual | 2.4139e+11 | 2,920 | 82669460.4 | Prob > F      | = | 0.0000  |
|          |            |       |            | R-squared     | = | 0.3822  |
|          |            |       |            | Adj R-squared | = | 0.3820  |
| Total    | 3.9071e+11 | 2,921 | 133758808  | Root MSE      | = | 9092.3  |

| MEANEAR~6YRS | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |          |
|--------------|-------------|-----------|-------|-------|----------------------|----------|
| AVGFACSAL    | .3281579    | .0077215  | 42.50 | 0.000 | .3130177             | .3432981 |
| _cons        | 12740.02    | 561.0436  | 22.71 | 0.000 | 11639.94             | 13840.11 |

## 1. Analysis of Variance (ANOVA) Table

ANOVA table shows variances used to calculate the model fit.

| Source   | SS         | df    | MS         |
|----------|------------|-------|------------|
| Model    | 1.4931e+11 | 1     | 1.4931e+11 |
| Residual | 2.4139e+11 | 2,920 | 82669460.4 |
| Total    | 3.9071e+11 | 2,921 | 133758808  |

Note 1.

$1.4931e + 11 = 1.4931 \times 10^{11} = 149,310,000,000$   
Stata uses 'e+number' to represent a very large number.

Note 2.  $n = 2,922$

**Source:** Source of variance

### Sum of Squares (SS)

- What is Explained SS (ESS)?
- What is Residual SS (RSS)?
- What is Total SS (TSS)?
- Then, what is  $R^2$ ?

### Degrees of freedom (df)

- df = the maximum number of independent values.
- Total variance has  $(n - 1)$  degrees of freedom
- Model df = number of independent variables (excluding the constant term)  $\equiv k$
- Residual df = Total df – Model df =  $n - k - 1$

### Mean Squares (MS) = SS/df

- MS are used to calculate the F-value.

$$F(k, n - k - 1) = \frac{MS_{Model}}{MS_{Residual}} = \frac{ESS/k}{RSS/(n - k - 1)}$$

## 2. Overall Model Fit

|               |   |                |
|---------------|---|----------------|
| Number of obs | = | <b>2,922</b>   |
| F(1, 2920)    | = | <b>1806.16</b> |
| Prob > F      | = | <b>0.0000</b>  |
| R-squared     | = | <b>0.3822</b>  |
| Adj R-squared | = | <b>0.3820</b>  |
| Root MSE      | = | <b>9092.3</b>  |

Which section should we refer to find the following information?

- Is the model significant?  
(Is this model at least better than running a regression without the set of independent variables I chose?)
  - $H_0$ : All coefficients are zero.
  - $H_a$ : At least one of the coefficients is not zero
- How well does the model explain the data (i.e., model fitness)?"

What is Root MSE (=standard error of the regression (SER))?

- SER measures the spread of observations around the regression line.
- A large SER indicates a wide scatter of data points around the regression line, suggesting that the actual values for many observations differ significantly from the predicted values.

### 3. Parameter Estimates

$$\text{Postgraduate earnings}_i = \beta_0 + \beta_1 \text{Faculty Salary}_i + u_i$$

|                           | MEANEAR~6YRS | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |          |
|---------------------------|--------------|-------------|-----------|-------|-------|----------------------|----------|
| y                         |              |             |           |       |       |                      |          |
| x                         | AVGFACSAL    | .3281579    | .0077215  | 42.50 | 0.000 | .3130177             | .3432981 |
| constant<br>(y-intercept) | _cons        | 12740.02    | 561.0436  | 22.71 | 0.000 | 11639.94             | 13840.11 |

Hypothesis Test  
on a parameter

A 95% Confidence Interval  
for a parameter

- What are the values of  $\widehat{\beta}_1$  and  $\widehat{\beta}_0$ ? How do we interpret them?
- How can we manually conduct a hypothesis test at the 5% significance level? ( $H_0: \beta_1 = 0, H_a: \beta_1 \neq 0,$ )
- Is  $\widehat{\beta}_1$  statistically significant at the 5% level? What about at the 1% level?
- A useful rule of thumb for quickly checking if an estimated coefficient is statistically significant at the 5% level
  - 1) Calculate the t-statistic:  $t = \text{coefficient}/\text{se}$ .
  - 2) If the absolute value of  $t$  is roughly greater than 2, the coefficient is significant at the 5% level.
    - When the t-statistic is 1.96, the p-value is 0.005. Therefore, if the t-stat is greater than 1.96, roughly 2, then the p-value is smaller than 5%. In this case, we can safely reject the null hypothesis that the parameter is zero.

# ECON108 Winter 2025

## Lab Assignment 3

Seeun Park

One important thing to keep in mind is that percentage points and percentages are different, and you should use the correct unit! A change in a percentage is measured in percentage points.

Percentage points (%p or pp)  $\neq$  Percentage

*Example:*

- The probability of getting a job with 3 interviews is 0.3 (or 30%).
- The probability of getting a job with 4 interviews is 0.35 (or 35%).
- If I currently have 3 interviews and get one more, my probability of getting a job increases by 5 **percentage points** (35% - 30% = **5pp**).

**Question 1.** What fraction of the sample support capital punishment for murder?

Answer: 58.55%

**. tab cappun**

| Favor or<br>oppose death<br>penalty for<br>murder | Freq.         | Percent       | Cum.          |
|---------------------------------------------------|---------------|---------------|---------------|
| Not applicable                                    | <b>7,500</b>  | <b>11.57</b>  | <b>11.57</b>  |
| Favor                                             | <b>37,946</b> | <b>58.55</b>  | <b>70.12</b>  |
| Oppose                                            | <b>15,604</b> | <b>24.08</b>  | <b>94.19</b>  |
| Don't know                                        | <b>3,410</b>  | <b>5.26</b>   | <b>99.45</b>  |
| No answer                                         | <b>354</b>    | <b>0.55</b>   | <b>100.00</b> |
| Total                                             | <b>64,814</b> | <b>100.00</b> |               |

**Question 2.** Convert the following categorical variables into indicator variables (see the do file for guidance)

Please check the attached do-file.

**Question 3.** Use the sum command to compare the characteristics of supporters of capital punishment vs. non-supporters. What is different about these two populations?

```
.
. su postlife_y white black other_race male female educ age if cappun_fav == 1
```

| Variable   | Obs    | Mean     | Std. dev. | Min | Max |
|------------|--------|----------|-----------|-----|-----|
| postlife_y | 27,226 | .8020642 | .3984508  | 0   | 1   |
| white      | 37,946 | .860117  | .3468702  | 0   | 1   |
| black      | 37,946 | .0923681 | .2895487  | 0   | 1   |
| other_race | 37,946 | .0475149 | .2127403  | 0   | 1   |
| male       | 37,946 | .4787329 | .4995541  | 0   | 1   |
| female     | 37,946 | .5212671 | .4995541  | 0   | 1   |
| educ       | 37,864 | 12.87579 | 2.965217  | 0   | 20  |
| age        | 37,833 | 46.14524 | 17.37598  | 18  | 89  |

```
. su postlife_y white black other_race male female educ age if cappun_fav == 0
```

| Variable   | Obs    | Mean     | Std. dev. | Min | Max |
|------------|--------|----------|-----------|-----|-----|
| postlife_y | 11,516 | .7911601 | .4064974  | 0   | 1   |
| white      | 15,604 | .6825814 | .4654868  | 0   | 1   |
| black      | 15,604 | .2446168 | .4298735  | 0   | 1   |
| other_race | 15,604 | .0728018 | .2598193  | 0   | 1   |
| male       | 15,604 | .3717637 | .4832912  | 0   | 1   |
| female     | 15,604 | .6282363 | .4832912  | 0   | 1   |
| educ       | 15,567 | 13.06816 | 3.500084  | 0   | 20  |
| age        | 15,549 | 45.45122 | 17.87997  | 18  | 89  |

- The share of postlife believers, Whites and males is larger, while the share of Blacks, individuals of other races, and females is smaller among supporters of capital punishment compared to non-supporters. However, the average years of education and age are similar between the two groups, although the average age of supporters is slightly older than that of non-supporters.

**Question 4.** Estimate the following regression:

$$\text{cappunfor} = \beta_0 + \beta_1 \text{postlifeyes} + \beta_2 \text{sexmale} + \beta_3 \text{racewhite} + \beta_4 \text{educ} + \beta_5 \text{age} + u$$

Interpret the estimated coefficients.

**. \*\*Q4**

**. reg cappun\_fav postlife\_y male white educ age**

| Source   | SS         | df     | MS         | Number of obs | = | 38,568 |
|----------|------------|--------|------------|---------------|---|--------|
| Model    | 420.494944 | 5      | 84.0989889 | F(5, 38562)   | = | 424.89 |
| Residual | 7632.68572 | 38,562 | .197932828 | Prob > F      | = | 0.0000 |
|          |            |        |            | R-squared     | = | 0.0522 |
|          |            |        |            | Adj R-squared | = | 0.0521 |
| Total    | 8053.18067 | 38,567 | .20881014  | Root MSE      | = | .4449  |

| cappun_fav | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|------------|-------------|-----------|-------|-------|----------------------|-----------|
| postlife_y | .0222165    | .0056722  | 3.92  | 0.000 | .0110989             | .0333341  |
| male       | .0875027    | .0045794  | 19.11 | 0.000 | .078527              | .0964785  |
| white      | .2373934    | .0057819  | 41.06 | 0.000 | .2260608             | .248726   |
| educ       | -.0067202   | .0007461  | -9.01 | 0.000 | -.0081825            | -.0052579 |
| age        | -.000425    | .0001324  | -3.21 | 0.001 | -.0006845            | -.0001654 |
| _cons      | .5626288    | .0135072  | 41.65 | 0.000 | .5361543             | .5891033  |

- $\beta_0$ : The probability that a person supports capital punishment when they do not believe in an afterlife, are female, non-white, have zero years of education, and are newborn (age = 0) is 0.56 (56%).  
(A newborn child, lol. Yes, this is yet another example of a meaningless y-intercept.)
- $\beta_1$ : The probability of supporting capital punishment is 2.22 percentage points (pp) higher for those who believe in an afterlife compared to those who do not, all else equal.
- $\beta_2$ : The probability of supporting capital punishment is 8.75pp higher for males compared to females, all else equal.
- $\beta_3$ : The probability of supporting capital punishment is 23.74pp higher for whites compared to non-whites, all else equal.
- $\beta_4$ : Each additional year of education decreases the probability of supporting capital punishment by 0.67pp, all else equal.
- $\beta_5$ : Each additional year of age decreases the probability of supporting capital punishment by 0.04pp, all else equal.
- All estimates are statistically significant at the 5% level.

**Question 5.** Predict the probability of supporting capital punishment for someone who believes in the afterlife, is female, is non-white, has 16 years of education, and is 40 years old.

Answer: 0.4603 (46.03%)

```
. **Q5
. margins, at(postlife_y=1 male = 0 white = 0 educ = 16 age = 40 )
```

Adjusted predictions  
Model VCE: OLS

Number of obs = 38,568

Expression: Linear prediction, predict()

```
At: postlife_y = 1
    male       = 0
    white      = 0
    educ       = 16
    age        = 40
```

|       | Delta-method |           |       |       |                      |
|-------|--------------|-----------|-------|-------|----------------------|
|       | Margin       | std. err. | t     | P> t  | [95% conf. interval] |
| _cons | .460324      | .0061526  | 74.82 | 0.000 | .4482648 .4723833    |

**Question 6.** What is the change in probability of support for capital punishment if the person in Q5 receives one more year of education?

Answer: -.0067 (-0.67pp)

```
. margins, dydx(educ) at(postlife_y=1 male = 0 white = 0 age = 40 )
```

Average marginal effects  
Model VCE: OLS

Number of obs = 38,568

Expression: Linear prediction, predict()

dy/dx wrt: educ

At: postlife\_y = 1  
male = 0  
white = 0  
age = 40

|      | Delta-method |           |       |       |                      |           |
|------|--------------|-----------|-------|-------|----------------------|-----------|
|      | dy/dx        | std. err. | t     | P> t  | [95% conf. interval] |           |
| educ | -.0067202    | .0007461  | -9.01 | 0.000 | -.0081825            | -.0052579 |

**Question 7.** Generate the predicted probability support for capital punishment for each observation in the dataset. How many predictions are out of the  $[0,1]$  range?

All predictions fall within the  $[0, 1]$  range.

```
. **Q7
. predict phat_lpm, xb
(22,450 missing values generated)
```

```
. su phat_lpm
```

| Variable | Obs    | Mean     | Std. dev. | Min      | Max      |
|----------|--------|----------|-----------|----------|----------|
| phat_lpm | 42,364 | .7023202 | .1047704  | .3916789 | .8923183 |

**Question 8.** What is the count R-squared? Do you think this model does a good job at predicting the probability of support for capital punishment?

**Answer:** 0.6411 “Yes, 64.11% of the sample was correctly predicted.”

```
.
. **Q8
. gen correct_lpm = 0

. replace correct_lpm = 1 if phat_lpm < 0.5 & cappun_fav == 0
(1,936 real changes made)

. replace correct_lpm = 1 if phat_lpm > 0.5 & cappun_fav == 1
(36,058 real changes made)

. replace correct_lpm = . if phat_lpm == .
(22,450 real changes made, 22,450 to missing)

.
. su correct_lpm
```

| Variable    | Obs    | Mean     | Std. dev. | Min | Max |
|-------------|--------|----------|-----------|-----|-----|
| correct_lpm | 42,364 | .6411104 | .4796804  | 0   | 1   |

**Question 9.** Instead of a linear probability model, estimate support for capital punishment using a probit regression:

$$P(\text{cappunfor} = 1) = \Phi(\beta_0 + \beta_1 \cdot \text{postlife\_y} + \beta_2 \cdot \text{sexmale} + \beta_3 \cdot \text{racewhite} + \beta_4 \cdot \text{educ} + \beta_5 \cdot \text{age})$$

Interpret the estimated coefficients.

```
. **Q9
. probit cappun_fav postlife_y male white educ age

Iteration 0:  Log likelihood = -23462.074
Iteration 1:  Log likelihood = -22489.975
Iteration 2:  Log likelihood = -22488.987
Iteration 3:  Log likelihood = -22488.987

Probit regression                               Number of obs =  38,568
                                                LR chi2(5)      = 1946.17
                                                Prob > chi2     =  0.0000
Log likelihood = -22488.987                    Pseudo R2      =  0.0415
```

| cappun_fav | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |          |
|------------|-------------|-----------|-------|-------|----------------------|----------|
| postlife_y | .0682034    | .017037   | 4.00  | 0.000 | .0348115             | .1015954 |
| male       | .2675692    | .0139475  | 19.18 | 0.000 | .2402326             | .2949059 |
| white      | .6510269    | .0166601  | 39.08 | 0.000 | .6183737             | .6836801 |
| educ       | -.0200533   | .002216   | -9.05 | 0.000 | -.0243966            | -.01571  |
| age        | -.0012538   | .0003989  | -3.14 | 0.002 | -.0020356            | -.000472 |
| _cons      | .1752168    | .0399807  | 4.38  | 0.000 | .0968561             | .2535775 |

- $\beta_0$ : The z-score of a person who does not believe in an afterlife, is female, non-white, has zero years of education, and is zero years old is 0.175.
- $\beta_1$ : The z-score for someone who believes in an afterlife is 0.068 higher than for a person who does not believe in an afterlife, all else equal.
- $\beta_2$ : The z-score for males is 0.268 higher than for females, all else equal.
- $\beta_3$ : The z-score for whites is 0.651 higher than for non-whites, all else equal.
- $\beta_4$ : For each additional year of education, the z-score decreases by 0.02, all else equal.
- $\beta_5$ : For each additional year of age, the z-score decreases by 0.001, all else equal.
- All estimates are statistically significant at the 5% level.

**Question 10.** Using your probit model estimates, predict the probability of supporting capital punishment for someone who believes in the afterlife, is female, is non-white, has 16 years of education, and is 40 years old.

Answer: 0.449 (44.9%)

```
. **Q10
. margins, at(postlife_y=1 male = 0 white = 0 educ = 16 age = 40 )

Adjusted predictions                                Number of obs = 38,568
Model VCE: OIM

Expression: Pr(cappun_fav), predict()
At: postlife_y = 1
    male        = 0
    white       = 0
    educ        = 16
    age         = 40
```

|       | Delta-method |           |       |       |                      |
|-------|--------------|-----------|-------|-------|----------------------|
|       | Margin       | std. err. | z     | P> z  | [95% conf. interval] |
| _cons | .4492389     | .0070147  | 64.04 | 0.000 | .4354902 .4629875    |

*Wait, what is the difference between the `margins` and `predict` commands?*

- **margins** calculates **average predicted values for groups of observations**. For example, the average predicted values for non-white, 40-year-old women with 16 years of education. Stata presents the results as a table.
- **predict** calculates **a predicted value for each observation** in the data, using the values of the explanatory variables for that observation. For example, it generates a predicted value for every individual in the dataset. To store these values, we create a new variable (e.g., `yhat`). To check the predicted values, we browse the dataset.

**Question 11.** What is the change in probability of support for capital punishment if the person in Q10 receives one more year of education?

Answer: -.008 (-0.8pp)

```
. *Q11
. margins, dydx(educ) at(postlife_y=1 male = 0 white = 0 age = 40 )

Average marginal effects                                     Number of obs = 38,568
Model VCE: OIM

Expression: Pr(cappun_fav), predict()
dy/dx wrt:  educ
At: postlife_y = 1
    male      = 0
    white     = 0
    age       = 40
```

|      | Delta-method |           | z     | P> z  | [95% conf. interval] |           |
|------|--------------|-----------|-------|-------|----------------------|-----------|
|      | dy/dx        | std. err. |       |       |                      |           |
| educ | -.0079662    | .0008763  | -9.09 | 0.000 | -.0096838            | -.0062487 |

*Comment:* I realized that the correct command for the probit model should include `educ = 16` inside `at()` because the model is nonlinear. When I add this, the estimated slope is -0.0079353, which is very close to -0.0079662. At age 12, the estimated effect is -0.0079912. Since the slope remains nearly constant across all age levels, this suggests an approximately linear relationship between age and the outcome variable.

**Question 12.** Generate the predicted probabilities (call it phat\_probit) and calculate the count R-squared.

**Answer: 0.6193** “61.93% of the sample was correctly predicted.”

```
. *Q12
. predict phat_probit, xb
(22450 missing values generated)

. su phat_probit
```

| Variable    | Obs    | Mean     | Std. dev. | Min       | Max      |
|-------------|--------|----------|-----------|-----------|----------|
| phat_probit | 42,364 | .5510599 | .2940306  | -.3336754 | 1.110611 |

```
.
.
. gen correct_probit = 0

. replace correct_probit = 1 if phat_probit < 0.5 & cappun_fav == 0
(4,734 real changes made)

. replace correct_probit = 1 if phat_probit > 0.5 & cappun_fav == 1
(32,334 real changes made)

. replace correct_probit = . if phat_probit == .
(22,450 real changes made, 22,450 to missing)

.
. su correct_probit
```

| Variable       | Obs    | Mean     | Std. dev. | Min | Max |
|----------------|--------|----------|-----------|-----|-----|
| correct_probit | 42,364 | .6192522 | .4855764  | 0   | 1   |

**Question 13.** Which model would you prefer? Why?

I prefer the linear probability model (LPM) because all predicted values lie within the range [0,1], and the count R-squared is higher than that of the probit model, indicating that the LPM has better predictability.